Test supermartingales are complete for analytic sequential hypotheses

Drona Khurana Rafael Frongillo

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Abstract

Composite e-processes provide a general framework for anytime-valid sequential testing, while composite test supermartingales form a more structured subclass whose validity can be verified through one-step conditional inequalities. In general, however, an e-process need not be a test supermartingale, and it is natural to ask when restricting attention to test supermartingales entails no loss of testing power.

We answer this question for a broad class of composite sequential hypotheses specified through admissible one-step conditional laws. After each finite history s, a correspondence $\mathcal{K}(s) \subseteq \Delta(\mathcal{Y})$ determines the distributions that may govern the next observation, with the local sets allowed to vary with the history and subject only to an analytic-graph regularity condition. The resulting null contains every path law generated by a universally measurable sequential selection from $\mathcal{K}$. We prove that every composite e-process valid for such an analytic sequential hypothesis is pointwise dominated by a Borel composite test supermartingale that is valid uniformly over the same null. Thus composite test supermartingales form a complete class for anytime-valid testing: arbitrary composite e-processes can be replaced, without loss of testing power, by composite test supermartingales.

Our result does not require local domination, convexity, or compactness of the admissible one-step sets, and includes examples not captured by previously studied finite-dimensional conditional-constraint models. We illustrate the scope of analytic sequential hypotheses via an array of examples.

1 Introduction

Anytime-valid inference allows evidence to be monitored continuously without fixing the stopping rule in advance (Howard et al., 2021; Ramdas et al., 2023; Grünwald et al., 2024). Composite e-processes provide a general way to express this guarantee: their expected value remains bounded at every stopping time under every law in the null. This definition is flexible, but global, since validity is a statement about all stopping rules under all null laws. Composite test supermartingales impose more structure, replacing this global requirement by a one-step conditional drift inequality at every history. That local form is useful both conceptually and operationally: the next move can be chosen from the data seen so far, and validity can be checked one step at a time. In a sequential clinical trial or online A/B test with binary outcomes, for example, an analyst may construct a test martingale by multiplying one-step likelihood-ratio factors, while choosing the next betting parameter predictably from the outcomes observed so far. Conditional on the past, the chosen factor has expectation one under the null, so this adaptation preserves validity one observation at a time. More generally, with a composite test supermartingale, it is enough to verify that the next-step factor has conditional expectation at most one under the null; one does not have to re-establish optional-stopping validity for the resulting process as a whole after each update. This local structure is one reason test supermartingales are especially convenient for constructing and optimizing sequential tests (Ville, 1939; Shafer et al., 2011; Ramdas et al., 2023). Every composite test supermartingale is a composite e-process, but the converse need not hold (Ramdas et al., 2022). For testing, however, the relevant issue is whether this extra generality can produce stronger threshold tests. Pointwise domination gives a direct power comparison: at any common rejection threshold, a dominating valid composite test supermartingale rejects on every path on which the composite e-process rejects, and no later, so its rejection probability is at least as large under every alternative law. We make this testing interpretation precise in § 2.4, after defining the complete-class property. This motivates a central question:

For which nulls is every valid composite e-process pointwise dominated by a valid composite test supermartingale?

A class for which this domination holds arises from a local specification of the null. After each finite history s, let $\mathcal{K}(s) \subseteq \Delta(\mathcal{Y})$ be the set of distributions allowed for the next observation. We assume that the graph of $\mathcal{K}$ is analytic and take the global null to contain every path law generated by a universally measurable sequential selection from these local sets. We call the resulting family an *analytic sequential hypothesis*. The formal local-to-global construction is given in § 2.2, with $\mathcal{K}$-selections defined in Definition 2.1 and the induced family in Definition 2.3. The local sets may vary with the history and need not be convex, compact, or dominated by a common reference measure. For every such analytic sequential hypothesis, we prove that every Borel composite e-process is pointwise dominated by a Borel composite test supermartingale valid for the same null.

Theorem 1.1 (Informal statement of Theorem 5.1)**.** *For every analytic sequential hypothesis, composite test supermartingales form a complete class for anytime-valid testing. Equivalently, every Borel composite e-process valid for the hypothesis is pointwise dominated by a Borel composite test supermartingale G valid for the same hypothesis, with $G_0 \leq 1$.*

The strength of the conclusion lies in the form of domination. Pointwise domination, formalized in Definition 2.6 and (3), is stronger than simultaneous almost-sure domination under the null: the dominating process is at least as large after every finite history, including histories that may have probability zero under every particular null law. Consequently, any threshold crossing of the original e-process is matched no later by the dominating test supermartingale. The formal statement is Theorem 5.1 in § 5. Figure 1 summarizes the main architecture of the result: local one-step constraints generate the sequential null, and global e-process validity is ultimately converted into historywise supermartingale inequalities.

Obtaining this historywise conclusion while returning to the Borel class is the main technical issue. The proof has two main steps. First, we construct a robust Snell envelope by backward recursion. The resulting envelope pointwise dominates the original e-process and satisfies the required one-step supermartingale inequalities, with initial value at most one. The finite- and infinite-horizon constructions are summarized in Protocols 1 and 2, respectively, with their properties established in Theorem 3.4. The construction entails optimization over an analytic graph, which preserves upper semianalyticity, but not necessarily Borel measurability. This first envelope may hence not be Borel adapted. To remedy this, we use a reflection argument to enlarge the process back into the Borel class without increasing its stopping-time cost (Proposition 4.2). Reflection however, can destroy the supermartingale property, so the two operations are alternated to obtain a non-decreasing sequence of alternating Borel-adapted, composite e-processes

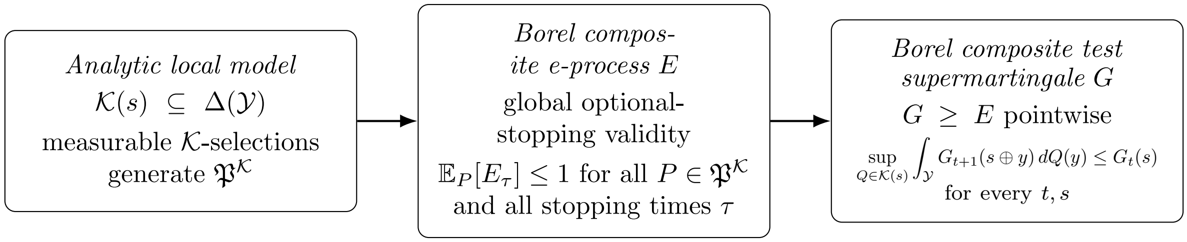

Figure 1: Architecture of the complete-class result. The analytic local model and universally measurable $\mathcal{K}$-selections (§ 2.2, Def. 2.1) generate the sequential hypothesis $\mathfrak{P}^{\mathcal{K}}$ (Def. 2.3). Global optional-stopping validity is the composite e-process condition (Def. 2.4, (1)). Theorem 5.1 replaces any such E by a Borel composite test supermartingale G (Def. 2.5) that pointwise dominates E (Def. 2.6). The construction proceeds through the `usa` Snell envelope (Thm. 3.4), a Borel reflection step (Prop. 4.2), and their interlacing (Prop. 4.3).

and upper semianalytic, composite supermartingales. Proposition 4.3 shows that their monotone limit is both Borel-adapted and a pointwise supermartingale. The measurability issue already appears in our bounded conditional mean running example, introduced in § 2.2 and revisited in lavender boxes throughout the paper. In § 3.1, we show that its local model, despite being compact and Borel, can nevertheless send a Borel continuation payoff to a non-Borel one-step value function. Example 6.5 gives a complementary case in which the local correspondence itself has an analytic, non-Borel graph. These two phenomena motivate the larger measurability class used in the proof. For readers less familiar with these regularity notions, Appendix C.1 gives a short guide to the Borel, analytic, and universally measurable classes and their function counterparts.

We next place the result in relation to existing complete-class results and sequential models.

1.1 Related work

The role of nonnegative martingales and supermartingales as canonical objects for anytime-valid inference has been studied in several nearby settings. Ramdas et al. (2020) show, in particular, that under a point null every e-process is dominated by a nonnegative martingale, and develop a broader admissibility theory for composite hypotheses under domination conditions. A closer analogue of our question appears in Ramdas et al. (2022): under local domination and fork-convexity of the null family, a composite Snell-envelope construction shows that every safe composite e-process is dominated by a composite test supermartingale valid under the whole null. Our result addresses the same complete-class question under a different structural regime. We work from an analytic correspondence of admissible one-step laws and include all path laws generated by measurable sequential selections. This allows genuinely nondominated models and does not require the local sets $\mathcal{K}(s)$ to be convex. Neither framework contains the other in general.

A complementary line of work starts from one-step distributional constraints. Larsson et al. (2026) develop a broad complete-class theory for e-variables under measurable constraints, including moment restrictions and classes such as sub-Gaussian distributions. Passing from one step to a sequential null requires the constraints and the corresponding e-variables to be chosen predictably across histories. Clerico (2026) carries this out for finitely many conditional constraints and shows that every composite e-process is pointwise dominated by a predictable product of affine one-step e-variables. For conditional means, Clerico (2025) gives an especially explicit version of this phenomenon. Our local formulation overlaps with this perspective at the level of one-step constraints, since many such restrictions can be encoded directly through the admissible sets $\mathcal{K}(s)$. These results are complementary in emphasis: the finite-dimensional theory yields an explicit affine characterization when it applies, while we seek a complete-class result for local models that may not admit any finite-feature representation. Related complete-class phenomena also appear in other formulations of anytime-valid inference. For example, Vovk and Wang (2024) show that martingale-based procedures form a complete class for merging sequential e-values. That problem takes sequential e-values themselves as the primitive objects, rather than a composite null over the observations, but it reinforces the recurring role of martingales as canonical representatives of anytime-valid procedures.

There is also a structural connection to rectangularity, pasting, and analytic product structure in robust decision theory and nondominated stochastic analysis. In finite event trees, the rectangularity of Epstein and Schneider (2003) has a local-product interpretation: the global family of priors contains all admissible branchwise recombinations of its one-step conditionals. Our formulation takes the local description as primitive and extends it to a general standard Borel observation space and infinite horizon, where branchwise choices must be made measurably. The analytic-graph assumption supplies the regularity needed for those selections. Thus our setup is best viewed as an analytically regular infinite-horizon local-product formulation, rather than as bare rectangularity itself.

A particularly close technical precedent is the analytic product structure used in nondominated mathematical finance and nonlinear expectations; see, e.g., Bouchard and Nutz (2015); Nutz and van Handel (2013). There too, global families of measures are built from local correspondences with analytic graphs and universally measurable selectors, usually for robust control, superhedging, or nonlinear dynamic programming. Here the same measurable structure is used to study whether an arbitrary anytime-valid process can be dominated by a single composite supermartingale with time-zero value at most one.

1.2 Examples and scope

In § 6, we develop five examples spanning familiar moment restrictions, nonparametric distributional constraints, and a genuinely analytic local model. Bounded conditional means serve as our running reference point: they admit familiar coin-betting constructions, while the local class is already nondominated (Howard et al., 2021; Clerico, 2025). Conditional first-order stochastic dominance and conditional sub-Gaussianity impose whole families of distributional or tail restrictions rather than finitely many conditional expectations (Arnold et al., 2026; Howard et al., 2021). Conditional Wasserstein balls give a different kind of distributional restriction, with the one-step upper expectation taking the familiar form of a Wasserstein distributionally robust optimization problem (Esfahani and Kuhn, 2018). Finally, a latent-feasibility extension of the bounded-mean model lets the active mean restriction depend on an analytic non-Borel compatibility set, connecting the measurable-selection issues in the proof to projection phenomena that also arise in partial identification (Kline and Tamer, 2016; Beresteanu et al., 2012; Gu and Russell, 2023).

These examples are meant to illustrate the scope of the local-model formulation rather than to supply a single canonical testing problem. Some fall outside earlier locally dominated settings Ramdas et al. (2022), some do not admit a finite-feature representation Clerico (2026), and the final example has a genuinely non-Borel local graph.

2 Sequential model and testing framework

We work on a canonical discrete-time path space. The composite null is specified locally, through the set of conditional laws that may govern the next observation after each finite history. We then introduce composite e-processes and composite test supermartingales for the resulting family of path laws, and formalize the complete-class question studied in this paper. For background on e-values, e-processes, and anytime-valid testing, see, e.g., Ramdas et al. (2023) and Grünwald et al. (2024).

2.1 Canonical path space

Let $\mathcal{Y}$ be a Polish space, equipped with its Borel σ-field $\mathcal{B}(\mathcal{Y})$. For $t \geq 0$, let $\mathcal{Y}^t$ denote the space of histories of length t, with $\mathcal{Y}^0 := \{\varnothing\}$. The *finite-history space* is $\mathsf{H} := \bigsqcup_{t\geq 0} \mathcal{Y}^t$. Since $\mathcal{Y}$ is Polish, each $\mathcal{Y}^t$ is Polish, and with the natural disjoint-union topology H is again Polish. For $s \in \mathcal{Y}^t$ and $y \in \mathcal{Y}$, we write $s \oplus y \in \mathcal{Y}^{t+1}$ for concatenation.

The *path space* is $\Omega := \mathcal{Y}^{\mathbb{N}}$, equipped with the product Borel σ-field $\mathcal{F} := \mathcal{B}(\Omega)$. Let $Y_t(\omega) := \omega_t$ denote the coordinate process, and write $Y_{1:t} := (Y_1, \ldots, Y_t)$. The *canonical filtration* is $\mathcal{F}_t := \sigma(Y_1, \ldots, Y_t)$ for $t \geq 1$, with $\mathcal{F}_0 := \{\varnothing, \Omega\}$.

Let $\Delta(\mathcal{Y})$ denote the space of Borel probability measures on $\mathcal{Y}$, equipped with the topology of weak convergence and its Borel σ-field. Since $\mathcal{Y}$ is Polish, $\Delta(\mathcal{Y})$ is Polish as well; this and the weak-topology facts used repeatedly below are recorded in Lemma B.3. For each $t \geq 0$, let $\mathcal{U}_t$ denote the *universal completion* of $\mathcal{B}(\mathcal{Y}^t)$, namely

$$\mathcal{U}_t := \bigcap_{Q\in\Delta(\mathcal{Y}^t)} \mathcal{B}(\mathcal{Y}^t)^Q,$$

where $\mathcal{B}(\mathcal{Y}^t)^Q$ denotes the Q-completion of $\mathcal{B}(\mathcal{Y}^t)$.

For orientation, the set classes used below satisfy Borel $\subseteq$ analytic $\subseteq$ universally measurable on a Polish space, while for extended-real-valued functions the corresponding implication is Borel measurable $\Longrightarrow$ upper semianalytic $\Longrightarrow$ universally measurable. Appendix C.1 gives the definitions and explains why projections and suprema in the robust Snell recursion naturally lead from the Borel class to the upper-semianalytic one.

Throughout, a nonnegative process $X = (X_t)_{t\geq 0}$ is identified with its history-space representatives $X_t : \mathcal{Y}^t \to [0, \infty]$, so that its canonical realization is $X_t(Y_{1:t})$. We call X *Borel adapted* if each X_t is Borel measurable. Larger measurability classes will arise naturally in our proofs and will be introduced when needed.

2.2 Local models and the induced sequential family

We describe the null locally by specifying which one-step distributions are admissible after each finite history. Let $\mathcal{K} : \mathsf{H} \rightrightarrows \Delta(\mathcal{Y})$ be a nonempty-valued correspondence. For a history $s \in \mathsf{H}$, the set $\mathcal{K}(s)$ is interpreted as the collection of admissible laws for the next observation following s.

Our basic regularity assumption is that the graph

$$\mathrm{Gr}(\mathcal{K}) := \big\{(s, Q) \in \mathsf{H} \times \Delta(\mathcal{Y}) : Q \in \mathcal{K}(s)\big\}$$

is *analytic*. We refer to such a correspondence as an *analytic local model*. Analytic-graph assumptions of this type are standard in universally measurable dynamic programming and in nondominated stochastic models; see, e.g., Bertsekas and Shreve (1996, Chapter 7), Nutz and van Handel (2013), and Bouchard and Nutz (2015). Strengthening the assumption to a Borel graph would not, by itself, make the Snell recursion Borel: even for Borel continuation values, taking the supremum over $Q \in \mathcal{K}(s)$ involves a projection of a Borel set, and such a projection need only be analytic. The bounded-mean construction in § 3.1 gives a concrete instance even for a compact, history-independent local set. Thus upper semianalyticity remains the general measurability guarantee without additional regularity; see Remark 3.5 after Theorem 3.4. Example 6.5 exhibits a different possibility: the local correspondence itself can have an analytic, genuinely non-Borel graph even though each of its values is a simple compact mean-constraint set. We do not assume that the sets $\mathcal{K}(s)$ are convex or compact, nor do we impose a common dominating measure.

A measurable choice of one admissible law at every history determines a probability law on the full path space.

Definition 2.1 ($\mathcal{K}$-selection)**.** *A* $\mathcal{K}$-selection *is a sequence $\kappa = (\kappa_t)_{t\geq 0}$ such that, for every $t \geq 0$, the map $\kappa_t : \mathcal{Y}^t \to \Delta(\mathcal{Y})$ is $\mathcal{U}_t/\mathcal{B}(\Delta(\mathcal{Y}))$-measurable and*

$$\kappa_t(s) \in \mathcal{K}(s) \qquad \textit{for every } s \in \mathcal{Y}^t.$$

For $A \in \mathcal{B}(\mathcal{Y})$, we write $\kappa_t(s, A) := \kappa_t(s)(A)$.

Since $\mathrm{Gr}(\mathcal{K})$ is analytic and its sections are nonempty, the Jankov–von Neumann theorem, stated as Theorem C.2 in Appendix C, guarantees the existence of a universally measurable selector. The use of universal rather than Borel measurability is important here: an analytic graph need not admit a Borel selector; the distinction and the near-maximizing form used later are recorded in Theorem C.2 and Proposition C.4.

Every $\mathcal{K}$-selection induces a path law. Lemma 2.2 makes the passage from the local specification $\mathcal{K}$ to a global family of probability measures precise.

Lemma 2.2 (Path law induced by a $\mathcal{K}$-selection)**.** *For every $\mathcal{K}$-selection $\kappa = (\kappa_t)_{t\geq 0}$, there exists a unique probability measure $P^\kappa \in \Delta(\Omega)$ such that, for every $n \geq 1$ and every bounded Borel function $f : \mathcal{Y}^n \to \mathbb{R}$,*

$$\mathbb{E}_{P^\kappa}\big[f(Y_1, \dots, Y_n)\big] = \int_{\mathcal{Y}} \cdots \int_{\mathcal{Y}} f(y_1, \dots, y_n)\, \kappa_{n-1}(y_{1:n-1}, dy_n) \cdots \kappa_1(y_1, dy_2)\kappa_0(\varnothing, dy_1).$$

Moreover, for every $t \geq 0$, $\kappa_t(Y_{1:t}, \cdot)$ is a version of the regular conditional law of Y_{t+1} given $\mathcal{F}_t$ under P^κ.

The proof is deferred to Appendix A. The two measure-theoretic ingredients are stated there explicitly: Lemma A.1 gives a Borel modification of a universally measurable kernel under a fixed marginal law, and Theorem A.2 is the Ionescu–Tulcea extension theorem for the resulting Borel kernels.

With the path law associated to each selection fixed, we can define the composite null generated by the local model.

Definition 2.3 (Analytic sequential hypothesis generated by $\mathcal{K}$)**.** *The* analytic sequential hypothesis generated by $\mathcal{K}$ *is*

$$\mathfrak{P}^{\mathcal{K}} := \{P^\kappa : \kappa \textit{ is a } \mathcal{K}\textit{-selection}\}.$$

When $\mathcal{K}$ is clear from context, we simply write $\mathfrak{P}$.

Thus $\mathfrak{P}^{\mathcal{K}}$ contains every path law obtained by selecting, at each finite history, an admissible next-step distribution from $\mathcal{K}(s)$ in a universally measurable way. In particular, admissible conditional laws may be recombined across histories without any additional global compatibility restriction beyond measurability.

Running example (bounded conditional mean).

A simple local model that we will return to throughout the paper is obtained by taking $\mathcal{Y} = [0, 1]$, fixing $\mu \in (0, 1)$, and setting

$$\mathcal{K}_\mu(s) := \left\{ Q \in \Delta([0, 1]) : \int y \, Q(dy) = \mu \right\}$$

for every history s. By Lemma B.3, the map $Q \mapsto \int y \, Q(dy)$ is continuous for weak convergence, because $y \mapsto y$ is bounded and continuous on $[0, 1]$. Hence $\mathcal{K}_\mu(s)$ is closed in the compact space $\Delta([0, 1])$, and is therefore compact; its graph is in fact Borel. A $\mathcal{K}_\mu$-selection may nevertheless choose a different mean-μ distribution after every history. Thus $\mathfrak{P}^{\mathcal{K}_\mu}$ is the family of path laws satisfying

$$\mathbb{E}_P[Y_{t+1} \mid \mathcal{F}_t] = \mu \qquad P\text{-almost surely for every } t,$$

with no independence or stationarity requirement on the remaining features of the conditional law. We refer to this as the *bounded conditional mean model.*

This local-to-global construction is the product structure behind the analytic sequential hypotheses studied here. Closely related product and pasting ideas appear under the names *rectangularity* and *analytic product structure* in robust control, multiple-priors models, and nondominated stochastic analysis (Epstein and Schneider, 2003; Iyengar, 2005; Nutz and van Handel, 2013; Bouchard and Nutz, 2015). The connection is discussed briefly in § 1.1.

The local specification itself is pointwise in the history. A $\mathcal{K}$-selection satisfies $\kappa_t(s) \in \mathcal{K}(s)$ at every $s \in \mathcal{Y}^t$, including histories that may have probability zero under the induced law P^κ. Once a particular path law is fixed, however, the identification of $\kappa_t(Y_{1:t}, \cdot)$ with a regular conditional distribution is necessarily only P^κ-almost sure.

2.3 Composite e-processes and composite test supermartingales

We now introduce the two classes of sequential testing processes that we compare. Our terminology follows the modern anytime-valid testing literature, in which composite e-processes encode validity under optional stopping and composite test supermartingales provide a betting-based sufficient construction (Shafer et al., 2011; Ramdas et al., 2023; Grünwald et al., 2024). Let $\mathcal{T}$ denote the collection of all stopping times with respect to the canonical filtration $(\mathcal{F}_t)_{t \geq 0}$, allowing the value ∞. For a nonnegative Borel-adapted process E, we adopt the standard convention

$$E_\infty := \limsup_{t \to \infty} E_t,$$

and use this convention in defining E_τ when τ may be infinite (Ramdas et al., 2022).

Definition 2.4 (Composite e-process)**.** *Let $\mathfrak{P} \subseteq \Delta(\Omega)$ be a composite null. A nonnegative Borel-adapted process $E = (E_t)_{t \geq 0}$ is a* $\mathfrak{P}$-e-process *if $E_0 \leq 1$ and*

$$\mathbb{E}_P[E_\tau] \leq 1 \qquad \textit{for every } P \in \mathfrak{P} \textit{ and every } \tau \in \mathcal{T}. \tag{1}$$

Thus a composite e-process remains valid under arbitrary optional stopping, including possibly infinite stopping times, uniformly over the null. An analyst may decide when to stop on the basis of the observations seen so far, while the expected stopped value remains bounded by one under every $P \in \mathfrak{P}$.

Test supermartingales satisfy the stronger requirement that the process have nonpositive conditional drift under every law in the null. For $P \in \Delta(\Omega)$, let $\mathcal{F}_t^P$ denote the P-completion of $\mathcal{F}_t$.

Definition 2.5 (Test supermartingale). *Let $\mathfrak{P} \subseteq \Delta(\Omega)$ be a composite null. A nonnegative Borel-adapted process $M = (M_t)_{t\geq 0}$ is a $\mathfrak{P}$-*test supermartingale *if $M_0 \leq 1$ and, for every $P \in \mathfrak{P}$ and $t \geq 0$,*

$$\mathbb{E}_P\left[M_{t+1} \,\middle|\, \mathcal{F}_t^P\right] \leq M_t \qquad P\text{-almost surely.} \tag{2}$$

The same process M must therefore satisfy (2) under every law in the composite null. Every composite $\mathfrak{P}$-test supermartingale is a composite $\mathfrak{P}$-e-process. Indeed, under each $P \in \mathfrak{P}$, nonnegative supermartingale convergence gives an almost-sure limit M_∞, and optional sampling applied to $\tau \wedge n$, followed by Fatou's lemma, yields

$$\mathbb{E}_P[M_\tau] \leq M_0 \leq 1$$

for every possibly infinite stopping time $\tau \in \mathcal{T}$ (see, e.g., Kallenberg, 2021). The converse need not hold. An e-process is constrained through its expected value at stopping times, but need not itself satisfy a one-step supermartingale inequality. A particularly sharp separation occurs for testing exchangeability, where nontrivial test supermartingales can be powerless even though powerful e-processes exist (Ramdas et al., 2022).

The question studied in this paper is whether this distinction matters for testing. Although composite e-processes form the more general class, it may still be possible to replace every valid composite e-process by a valid composite test supermartingale that is at least as large after every possible history.

2.4 Pointwise domination and the complete-class property

We formalize this comparison using a pathwise notion of domination.

Definition 2.6 (Pointwise domination). *Let $E = (E_t)_{t\geq 0}$ and $M = (M_t)_{t\geq 0}$ be adapted processes. We say that M* pointwise dominates, *or* superreplicates, *E if*

$$M_t(s) \geq E_t(s) \qquad \textit{for every } t \geq 0 \textit{ and every } s \in \mathcal{Y}^t. \tag{3}$$

The historywise requirement (3) is stronger than requiring domination P-almost surely for every $P \in \mathfrak{P}$. The inequality must hold at every finite history, including histories that are null under some or all laws in the composite null. This pathwise order induces the complete-class notion in Definition 2.7.

Definition 2.7 (Complete class). *Let $\mathfrak{P} \subseteq \Delta(\Omega)$. We say that* composite test supermartingales form a complete class *for $\mathfrak{P}$ if every $\mathfrak{P}$-e-process E is pointwise dominated by some $\mathfrak{P}$-test supermartingale M.*

The complete-class property gives a direct testing interpretation. Fix a confidence level $\alpha \in (0, 1)$ and let

$$\tau_E := \inf\left\{t \geq 0 : E_t \geq \frac{1}{\alpha}\right\}$$

be the rejection time obtained by thresholding a composite e-process. If M pointwise dominates E and τ_M denotes the analogous rejection time for M, then

$$\tau_M \le \tau_E$$

along every path. Moreover, since M is a nonnegative composite test supermartingale, Ville's inequality (Ville, 1939) gives, for every $P \in \mathfrak{P}$,

$$P(\tau_M < \infty) = P\left(\sup_{t\ge 0} M_t \ge \frac{1}{\alpha}\right) \le \alpha M_0 \le \alpha$$

Thus thresholding M at $1/\alpha$ yields an anytime-valid level-α test, while pointwise domination ensures that any rejection made by E is made no later by M. In this sense, restricting attention to composite test supermartingales entails no loss of anytime-valid testing power.

Theorem 5.1 establishes this complete-class property for the analytic sequential hypotheses $\mathfrak{P}^{\mathcal{K}}$ generated by Definition 2.3.

Remark 2.8 (Measurability convention)**.** The testing objects in Definitions 2.4 and 2.5 are Borel adapted. The analytic-graph assumption on $\mathcal{K}$, however, naturally leads to universally measurable selectors and, in the Snell-envelope construction used in the proof, to upper semianalytic and universally adapted intermediate processes. We introduce these larger measurability classes only when they become necessary. § 4, culminating in Proposition 4.3, passes from this universally measurable intermediate construction back to a Borel-adapted dominating composite test supermartingale.

3 Constructing the Snell envelope

The first step in the proof is to show that a composite e-process can be enlarged to a process satisfying the supermartingale inequalities pointwise at every history. The natural construction is a robust Snell envelope. For a fixed probability law, the classical Snell envelope is the smallest supermartingale dominating a reward process and is the standard dynamic-programming object for optimal stopping (Peskir and Shiryaev, 2006; Shiryaev, 2008). A composite version also appears in Ramdas et al. (2022, Theorem 14): for a locally dominated, fork-convex family of laws, they construct the smallest composite nonnegative supermartingale dominating a safe composite e-process by taking essential suprema of conditional stopped rewards relative to a common reference measure.

Our construction serves the same basic purpose, of converting optional-stopping safety into dynamic supermartingale inequalities, but is organized differently. The null is given locally through the historywise sets $\mathcal{K}(s)$, so the recursion is performed pointwise on the history space using a robust one-step expectation. No common locally dominating reference measure is required, and we do not assume fork-convexity of the global family (Ramdas et al., 2022). The tradeoff is measurability: the one-step upper expectation used in the Snell recursion, defined in (4), preserves upper semianalyticity rather than Borel measurability, so a single Snell step may leave the Borel class. We also do not need to identify an almost-surely smallest composite envelope: for the complete-class theorem, it is enough to construct a pointwise dominating process satisfying the local inequalities and having time-zero value at most one.

Because the one-step optimization may leave the Borel class, we first enlarge the measurability class in which the intermediate recursion is carried out. A function f on a Polish space is called *upper semianalytic* (`usa`) if $\{f > a\}$ is analytic for every $a \in \mathbb{R}$. Every `usa` function is universally measurable by Lemma C.1; Appendix C.1 summarizes how this class sits between Borel and universal measurability. We call a nonnegative process $X = (X_t)_{t\geq 0}$ a ***usa** process* if each history-space representative $X_t : \mathcal{Y}^t \to [0, \infty]$ is usa. A bounded stopping time τ is called *universally measurable* if, for every t, the event $\{\tau \leq t\}$ can be written as $\{Y_{1:t} \in A_t\}$ for some $A_t \in \mathcal{U}_t$. We write $\mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$ for the class of bounded universally measurable stopping times.

The e-process definition in § 2.3 quantifies over all stopping times. The finite-horizon Snell arguments below require only the bounded universally measurable subclass. Accordingly, for the intermediate arguments in this section, we call a nonnegative usa process E *bounded-stopping valid for* $\mathfrak{P}^{\mathcal{K}}$ if $E_0 \leq 1$ and $\mathbb{E}_{P^\kappa}[E_\tau] \leq 1$ for every $\mathcal{K}$-selection κ and every $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. Lemma 3.1 lets us pass from a Borel composite e-process to this intermediate validity condition.

Lemma 3.1 (Borel composite e-processes are bounded-stopping valid)**.** *Every Borel $\mathfrak{P}^{\mathcal{K}}$-e-process is bounded-stopping valid for $\mathfrak{P}^{\mathcal{K}}$.*

Proof. Fix a $\mathcal{K}$-selection κ and $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. By Lemma B.1, there is a bounded Borel stopping time σ such that $\sigma = \tau$ P^κ-almost surely. Since E is Borel adapted, $E_\sigma = E_\tau$ P^κ-almost surely, and Definition 2.4 gives $\mathbb{E}_{P^\kappa}[E_\tau] = \mathbb{E}_{P^\kappa}[E_\sigma] \leq 1$. Every Borel function is usa, so the claim follows. □

3.1 The one-step upper expectation and finite-horizon recursion

For a nonnegative usa function $f : \mathcal{Y}^{t+1} \to [0, \infty]$, define the *one-step upper expectation* $\mathsf{T}_t f : \mathcal{Y}^t \to [0, \infty]$ by

$$(\mathsf{T}_t f)(s) := \sup_{Q \in \mathcal{K}(s)} \int_{\mathcal{Y}} f(s \oplus y)\, Q(dy). \tag{4}$$

Thus the quantity in (4) is the largest expected continuation value compatible with the local model after history s. The analytic-graph assumption ensures that this operation preserves the measurability class needed for the recursion.

Lemma 3.2 (Analytic dynamic-programming facts)**.** *Let $f, f_n : \mathcal{Y}^{t+1} \to [0, \infty]$ be **usa**. Then $\mathsf{T}_t f$ is **usa**, T_t is monotone, and if $f_n \uparrow f$ pointwise then $\mathsf{T}_t f_n \uparrow \mathsf{T}_t f$ pointwise. Moreover, if f is bounded, then for every $\delta > 0$ there is a universally measurable selector $\kappa^\delta_t : \mathcal{Y}^t \to \Delta(\mathcal{Y})$ satisfying $\kappa^\delta_t(s) \in \mathcal{K}(s)$ and*

$$\int_{\mathcal{Y}} f(s \oplus y)\, \kappa^\delta_t(s, dy) \geq (\mathsf{T}_t f)(s) - \delta$$

for every $s \in \mathcal{Y}^t$.

The descriptive-set-theoretic ingredients are stated separately in Appendix C: Lemma C.3 treats integration of a usa continuation value against a variable law, Lemma C.1 records the projection and universal-measurability facts, and Proposition C.4 supplies universally measurable near-maximizers. The proof of Lemma 3.2 is then a direct application of those results. Related nondominated dynamic-programming constructions appear in Nutz and van Handel (2013); Bouchard and Nutz (2015). The final statement is the only place where a maximizing conditional law is needed; importantly, the supremum defining T_t need not be attained, so the selector is only approximately maximizing.

Running example (continued): bounded conditional mean.

For the bounded conditional mean model $\mathcal{K}_\mu$ introduced in the running example, fix a history s and write $f_s(y) := f(s \oplus y)$. If f_s is bounded and continuous, then

$$(\mathsf{T}_t f)(s) = \sup_{Q:\, \mathbb{E}_Q Y = \mu} \mathbb{E}_Q[f_s(Y)] = \operatorname{conc}(f_s)(\mu),$$

where $\operatorname{conc}(f_s)$ denotes the least concave majorant of f_s on $[0,1]$. Indeed, every concave majorant $g \geq f_s$ gives

$$\mathbb{E}_Q[f_s(Y)] \leq \mathbb{E}_Q[g(Y)] \leq g(\mathbb{E}_Q Y) = g(\mu),$$

and in one dimension the bound is attained, or approximated arbitrarily well, by a distribution supported on at most two points whose barycenter is μ.

When f is jointly continuous, compactness of $\mathcal{K}_\mu$ and the maximum theorem show that $s \mapsto (\mathsf{T}_t f)(s)$ is continuous. Finite-horizon Snell envelopes of continuous reward processes are therefore continuous, and in particular Borel. This regularity should not be confused with a general Borel-preservation statement: for an arbitrary Borel continuation payoff, the robust supremum need only be upper semianalytic.

A Borel payoff with a non-Borel Snell step. The distinction is genuine even for this compact Borel model. Adapting Frongillo (2025, Example 4.12), let $B \subseteq [0,1]^2$ be Borel with analytic non-Borel projection $A := \pi_1(B)$, whose existence follows from Suslin's classical theorem (Kechris, 1995, Section 14). Define the continuous map

$$h_\mu(y) := \begin{cases} (\mu - y)/\mu, & y \leq \mu, \\ (y-\mu)/(1-\mu), & y \geq \mu. \end{cases}$$

The terminal payoff

$$X(y_1, y_2) := \mathbb{1}\big\{(y_1, h_\mu(y_2)) \in B\big\}$$

is Borel. For $a \in [0,1]$, set

$$y_a^- := \mu(1-a), \qquad y_a^+ := \mu + (1-\mu)a, \qquad Q_a := (1-\mu)\delta_{y_a^-} + \mu\delta_{y_a^+}.$$

Then $Q_a \in \mathcal{K}_\mu$, while $h_\mu(y_a^-) = h_\mu(y_a^+) = a$. Consequently,

$$(\mathsf{T}_1 X)(y_1) = \sup_{Q:\, \mathbb{E}_Q Y = \mu} \mathbb{E}_Q[X(y_1, Y)] = \mathbb{1}_A(y_1).$$

Indeed, the value is zero when $y_1 \notin A$, while for $y_1 \in A$ one can choose a with $(y_1, a) \in B$ and then use Q_a to attain value one. Thus a Borel continuation payoff can produce a non-Borel one-step upper expectation; taking zero reward before the terminal time gives a finite-horizon Snell envelope whose time-1 value is non-Borel. This substantiates why the larger measurability class used in § 3 is needed even when the local correspondence itself is Borel.

This pathology concerns the Snell envelope of general Borel rewards. The familiar bounded-mean betting martingales behave much more regularly and fit the recursion directly. If

$$M_{t+1}(s \oplus y) = M_t(s)\big(1 + \lambda_t(s)(y - \mu)\big)$$

Protocol 1: Finite-horizon robust Snell envelope

Input: A horizon $N \in \mathbb{N}$ and a nonnegative `usa` process $E = (E_t)_{t=0}^N$.
Output: The finite-horizon robust Snell envelope $M^{(N)} = (M_t^{(N)})_{t=0}^N$.
Set $M_N^{(N)} := E_N$.
for $t = N-1, \ldots, 0$ **do**
  Define $M_t^{(N)} : \mathcal{Y}^t \to [0, \infty]$ by

$$M_t^{(N)}(s) := \max \left\{ \underbrace{E_t(s)}_{\text{stop}}, \underbrace{(\mathsf{T}_t M_{t+1}^{(N)})(s)}_{\text{continue}} \right\}, \qquad s \in \mathcal{Y}^t.$$

with a predictable stake chosen so that the factor is nonnegative, then

$$(\mathsf{T}_t M_{t+1})(s) = M_t(s).$$

Thus these classical betting processes are fixed points of the one-step Snell operation; the Snell envelope of a general composite e-process need not itself have this affine betting form.

With the one-step operator in hand, we now build the envelope over a finite horizon. Fix a finite horizon $N \in \mathbb{N}$ and a nonnegative `usa` process E. The finite-horizon robust Snell envelope is obtained by the backward recursion in Protocol 1. At each history, the envelope compares the value $E_t(s)$ obtained by stopping immediately with the largest expected value of continuing for one more step under an admissible conditional law.

By Lemma 3.2, every $M_t^{(N)}$ produced by Protocol 1 is `usa`. Moreover, the two defining properties of the envelope are immediate from the recursion:

$$M_t^{(N)} \geq E_t, \qquad \mathsf{T}_t M_{t+1}^{(N)} \leq M_t^{(N)} \tag{5}$$

pointwise on $\mathcal{Y}^t$. The first inequality in (5) gives domination of the original process, while the second is the robust one-step supermartingale inequality. The remaining issue is to show that the envelope still satisfies $M_0^{(N)} \leq 1$. Proposition 3.3 establishes precisely this bound.

3.2 Finite-horizon initial-value bound

The key argument is most transparent for bounded rewards. The envelope prescribes stopping when it first meets the reward, while measurable selection supplies conditional laws that come arbitrarily close to realizing the worst-case continuation value.

Proposition 3.3 (Finite-horizon initial-value bound)**.** *Let E be a nonnegative `usa` process that is bounded-stopping valid for $\mathfrak{P}^{\mathcal{K}}$, and let $M^{(N)}$ be its finite-horizon robust Snell envelope. Then $M_0^{(N)} \leq 1$ for every $N \in \mathbb{N}$. More generally, if a nonnegative `usa` process E satisfies*

$$\mathbb{E}_{P^\kappa}[E_\tau] \leq c$$

for every $\mathcal{K}$-selection κ and every $\tau \in \mathcal{T}_{\text{bd}}^{\text{u}}$, then $M_0^{(N)} \leq c$ for every N.

Proof. We first suppose that $E_0, \dots, E_N$ are bounded by a common constant $L < \infty$. Then the recursion and preservation of constants by T_t imply $0 \le M_t^{(N)} \le L$ for every $t \le N$. Fix $\varepsilon > 0$, and choose $\delta_0, \dots, \delta_{N-1} > 0$ with $\sum_{t=0}^{N-1} \delta_t < \varepsilon$. For each $t < N$, Lemma 3.2 gives a universally measurable selector κ_t^ε such that

$$\int_{\mathcal{Y}} M_{t+1}^{(N)}(s \oplus y)\, \kappa_t^\varepsilon(s, dy) \ge (\mathsf{T}_t M_{t+1}^{(N)})(s) - \delta_t$$

for every $s \in \mathcal{Y}^t$. Complete these kernels after time N using any universally measurable selector of $\mathcal{K}$ supplied by Theorem C.2, and let P^{κ^ε} be the induced path law.

Define the first contact time

$$\tau^* := \inf\bigl\{0 \le t \le N : M_t^{(N)}(Y_{1:t}) = E_t(Y_{1:t})\bigr\}.$$

Since $M_N^{(N)} = E_N$, this stopping time is finite, and Lemma B.5 gives $\tau^* \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. On $\{t < \tau^*\}$ we have $M_t^{(N)} > E_t$, and hence the Snell recursion gives $M_t^{(N)} = \mathsf{T}_t M_{t+1}^{(N)}$. Using Lemma B.4 and the choice of κ_t^ε, we therefore obtain

$$\mathbb{E}_{P^{\kappa^\varepsilon}}\left[M_{t+1}^{(N)}(Y_{1:t+1}) \,\middle|\, \mathcal{F}_t^{P^{\kappa^\varepsilon}} \right] \ge M_t^{(N)}(Y_{1:t}) - \delta_t$$

on $\{t < \tau^*\}$, P^{κ^ε}-almost surely. Consequently,

$$Z_t := M_{t\wedge\tau^*}^{(N)}(Y_{1:t\wedge\tau^*}) + \sum_{j=0}^{(t\wedge\tau^*)-1} \delta_j, \qquad 0 \le t \le N,$$

is a bounded P^{κ^ε}-submartingale. At the stopping time, exact contact gives $M_{\tau^*}^{(N)} = E_{\tau^*}$, and hence

$$M_0^{(N)} = Z_0 \le \mathbb{E}_{P^{\kappa^\varepsilon}}[Z_N] \le \mathbb{E}_{P^{\kappa^\varepsilon}}[E_{\tau^*}] + \sum_{t=0}^{N-1} \delta_t < 1 + \varepsilon.$$

The bounded-stopping validity condition applies because τ^* is a bounded universally measurable stopping time. Letting $\varepsilon \downarrow 0$ yields $M_0^{(N)} \le 1$ in the bounded case.

For a general nonnegative E, set $E_t^{(m)} := E_t \wedge m$ and let $M^{(N,m)}$ be the corresponding finite-horizon envelope. Each $E^{(m)}$ is again bounded-stopping valid for $\mathfrak{P}^{\mathcal{K}}$, so the bounded argument gives $M_0^{(N,m)} \le 1$. By continuity from below of T_t and backward induction, $M_t^{(N,m)} \uparrow M_t^{(N)}$ for every $t \le N$. Therefore $M_0^{(N)} = \sup_m M_0^{(N,m)} \le 1$. The general bound with right-hand side c follows from the same argument, with the e-process bound 1 replaced throughout by c. □

3.3 Infinite-horizon Snell domination

We now let the stopping horizon increase. Protocol 2 records the construction: compute the finite-horizon envelopes from Protocol 1 and pass to their pointwise monotone limit.

Protocol 2: Infinite-horizon robust Snell envelope

Input: A nonnegative `usa` process $E = (E_t)_{t\ge 0}$.
Output: The infinite-horizon robust Snell envelope $M = (M_t)_{t\ge 0}$.
For each horizon $N \in \mathbb{N}$, construct the finite-horizon envelope $M^{(N)}$ using Protocol 1.
For every $t \ge 0$, set

$$M_t := \sup_{N\ge t} M_t^{(N)} = \lim_{N\to\infty} M_t^{(N)},$$

where the limit is pointwise on $\mathcal{Y}^t$.

The finite-horizon envelopes are monotone in the horizon: for fixed t, $M_t^{(N)} \le M_t^{(N+1)}$ whenever $N \ge t$. Indeed, $M_N^{(N+1)} = E_N \vee \mathsf{T}_N E_{N+1} \ge E_N = M_N^{(N)}$, and monotonicity of the one-step operators propagates this inequality backwards. Thus the limit in Protocol 2 is well defined and retains the pointwise domination and one-step inequalities established at finite horizons, while its time-zero value remains at most one.

Theorem 3.4 (`usa` Snell domination)**.** *Let E be a nonnegative `usa` process that is bounded-stopping valid for $\mathfrak{P}^{\mathcal{K}}$. Then the process $M = (M_t)_{t\ge 0}$ constructed in Protocol 2 is `usa`, satisfies $M_t \ge E_t$ pointwise for every t, has $M_0 \le 1$, and obeys*

$$\mathsf{T}_t M_{t+1}(s) \le M_t(s) \qquad \text{for every } t \ge 0 \text{ and } s \in \mathcal{Y}^t.$$

Consequently, for every $\mathcal{K}$-selection κ, the canonical process $M_t(Y_{1:t})$ is a nonnegative P^κ-supermartingale with respect to the P^κ-completed canonical filtration.

More generally, if a nonnegative `usa` process E satisfies

$$\mathbb{E}_{P^\kappa}[E_\tau] \le c$$

for every $\mathcal{K}$-selection κ and every $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$, then the same construction satisfies $M_0 \le c$.

Proof. For each fixed t, M_t is the increasing countable supremum of the `usa` functions $M_t^{(N)}$, and is therefore `usa`. Moreover, $M_t \ge M_t^{(t)} = E_t$ pointwise, while Proposition 3.3 gives $M_0 = \sup_N M_0^{(N)} \le 1$. Finally, continuity from below of T_t yields

$$\mathsf{T}_t M_{t+1} = \sup_{N \ge t+1} \mathsf{T}_t M_{t+1}^{(N)} \le \sup_{N \ge t+1} M_t^{(N)} \le M_t.$$

Fixing a $\mathcal{K}$-selection κ, Lemma B.4 converts this pointwise kernel inequality into

$$\mathbb{E}_{P^\kappa}\left[M_{t+1}(Y_{1:t+1}) \,\middle|\, \mathcal{F}_t^{P^\kappa}\right] \le M_t(Y_{1:t}) \qquad P^\kappa\text{-almost surely,}$$

which is the required supermartingale property.

For the final statement, the scaled part of Proposition 3.3 gives $M_0^{(N)} \le c$ for every N. Hence

$$M_0 = \sup_N M_0^{(N)} \le c.$$

□

Remark 3.5 (Borel graphs do not suffice for Borel Snell envelopes)**.** The upper semianalytic conclusion of Theorem 3.4 is not merely a consequence of allowing $\mathrm{Gr}(\mathcal{K})$ to be analytic rather than Borel. Even if $\mathrm{Gr}(\mathcal{K})$ is Borel, the upper transition operator need not preserve Borel measurability for arbitrary Borel continuation values: its strict superlevel sets arise as projections of Borel subsets of $\mathcal{Y}^t \times \Delta(\mathcal{Y})$, and such projections need only be analytic. Consequently, restricting the local model to Borel graphs would not by itself remove the need for the Borelization step of § 4. Borel or continuous Snell envelopes in more regular examples below rely on additional structure, such as compactness and continuity, rather than on the Borel-graph property alone.

The infinite-horizon construction in Protocol 2, formalized by Theorem 3.4, is the first domination step in the proof of our main result. Starting from a Borel composite e-process for $\mathfrak{P}^{\mathcal{K}}$, Lemma 3.1 lets us apply the theorem and obtain a pointwise dominating supermartingale M with $M_0 \le 1$. The construction also satisfies the stronger one-step inequalities at every finite history. What it does not yet provide is Borel measurability: even when E is Borel, the operator T_t need only preserve upper semianalyticity. § 4 addresses precisely this remaining obstruction.

4 From universal measurability to Borel supermartingales

The Snell construction of § 3, summarized by Protocol 2, produces the desired pointwise one-step inequalities, but generally only within the upper-semianalytic class. A second operation is therefore needed to return to Borel processes. The key observation is that a `usa` process whose stopped expectations are bounded by its time-zero value can be enlarged pointwise to a Borel process with the same optional-stopping guarantee and the same time-zero value. This reflection step need not preserve the one-step supermartingale inequalities, so we alternate it with the Snell construction and then pass to a monotone limit.

The argument produces two interlaced sequences. The reflected processes provide Borel adaptedness, while the intervening Snell envelopes provide the stronger pointwise supermartingale inequalities. Neither sequence alone has both properties after a single step. The monotone interlacing is what allows the limit to inherit Borel measurability from the reflected sequence and the local supermartingale structure from the Snell sequence.

4.1 Reflecting a `usa` process into the Borel class

The reflection step is most conveniently formulated at the level of analytic subgraphs. For an analytic set $D \subseteq \mathsf{H} \times [0, \infty)$, write $D_s := \{u \geq 0 : (s, u) \in D\}$ and let $g_D(s) := \lambda(D_s)$, where λ denotes Lebesgue measure on $[0, \infty)$. Define its *robust stopped-value cost* by

$$\mathcal{C}(D) := \sup_{\kappa} \sup_{\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}} \mathbb{E}_{P^\kappa}\big[g_D(Y_{1:\tau})\big],$$

where the first supremum ranges over all $\mathcal{K}$-selections. The functional $\mathcal{C}$ is monotone under set inclusion. The descriptive-set-theoretic input for reflection is that an analytic set can be enlarged to a Borel set without increasing this cost.

Proposition 4.1 (Borel hull preserving stopped-value cost)**.** *Let $D \subseteq \mathsf{H} \times [0, \infty)$ be analytic with $\mathcal{C}(D) < \infty$. Then there exists a Borel set $B \subseteq \mathsf{H} \times [0, \infty)$ such that $D \subseteq B$ and $\mathcal{C}(B) = \mathcal{C}(D)$.*

The proof is given in Appendix D. It combines a parameterized version of the finite-horizon dynamic-programming argument from § 3 with the dual First Reflection Theorem from descriptive set theory (Kechris, 1995, Theorem 35.10 and the discussion following it). We use Proposition 4.1 through its process-level consequence, Proposition 4.2.

Proposition 4.2 (Process reflection)**.** *Let $X = (X_t)_{t \geq 0}$ be a nonnegative* `usa` *process with $X_0 < \infty$, and suppose that*

$$\mathbb{E}_{P^\kappa}[X_\tau] \leq X_0$$

for every $\mathcal{K}$-selection κ and every $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. Then there exists a nonnegative Borel-adapted process $\mathsf{R}(X)$ such that $\mathsf{R}(X)_t(s) \geq X_t(s)$ for every t and $s \in \mathcal{Y}^t$, $\mathsf{R}(X)_0 = X_0$, and

$$\mathbb{E}_{P^\kappa}[\mathsf{R}(X)_\tau] \leq \mathsf{R}(X)_0$$

for every $\mathcal{K}$-selection κ and every $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$.

Proof. View X as a function on the disjoint-union history space H and define its strict subgraph

$$D_X := \{(s, u) \in \mathsf{H} \times [0, \infty) : 0 \leq u < X(s)\}.$$

Since X is `usa`, D_X is analytic, and its section length satisfies $g_{D_X}(s) = X(s)$ for every history s. The assumed optional-stopping bound gives $\mathcal{C}(D_X) \leq X_0$, while the deterministic stopping time $\tau \equiv 0$ gives the reverse inequality. Hence $\mathcal{C}(D_X) = X_0$.

By Proposition 4.1, there is a Borel set $B \supseteq D_X$ with $\mathcal{C}(B) = X_0$. Define $\mathsf{R}(X)(s) := \lambda(B_s)$. Because B is Borel, $\mathsf{R}(X)$ is Borel adapted, and $B \supseteq D_X$ gives $\mathsf{R}(X) \geq X$ pointwise. Moreover, stopping at time zero gives $\mathsf{R}(X)_0 \leq \mathcal{C}(B) = X_0$, while pointwise domination gives $\mathsf{R}(X)_0 \geq X_0$. Thus $\mathsf{R}(X)_0 = X_0$. Finally, the definition of $\mathcal{C}(B)$ gives $\mathbb{E}_{P^\kappa}[\mathsf{R}(X)_\tau] \leq \mathcal{C}(B) = \mathsf{R}(X)_0$ for every κ and every $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. □

The reflection proposition applies in particular to the output of the Snell construction. Indeed, if M is a nonnegative `usa` process with $M_0 < \infty$ satisfying $\mathsf{T}_t M_{t+1} \leq M_t$ pointwise, then Theorem 3.4 and bounded optional sampling imply

$$\mathbb{E}_{P^\kappa}[M_\tau] \leq M_0$$

for every $\mathcal{K}$-selection κ and every $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. Thus reflection restores Borel measurability while leaving the time-zero value of the `usa` Snell envelope unchanged. It does not, however, necessarily preserve the local inequalities $\mathsf{T}_t M_{t+1} \leq M_t$. This is why one further Snell step is required.

4.2 Alternating Snell and reflection

We now combine the Snell operation from Protocol 2 and Theorem 3.4 with the reflection operation of Proposition 4.2. Starting from a Borel composite e-process, the Snell construction produces a dominating `usa` process satisfying the pointwise one-step inequalities; reflection then produces a still larger Borel process whose stopped expectations are bounded by its own time-zero value at every bounded universally measurable stopping time. Repeating this procedure produces the increasing sequence formalized in Proposition 4.3, whose limit is simultaneously Borel and a pointwise supermartingale.

Proposition 4.3 (Alternating Snell–reflection construction)**.** *Let E be a Borel $\mathfrak{P}^{\mathcal{K}}$-e-process. Then there exists a nonnegative Borel-adapted process $G = (G_t)_{t\geq 0}$ such that $G_0 \leq 1$, $G_t(s) \geq E_t(s)$ for every $t \geq 0$ and $s \in \mathcal{Y}^t$, and*

$$\mathsf{T}_t G_{t+1}(s) \leq G_t(s) \qquad \text{for every } t \geq 0 \text{ and } s \in \mathcal{Y}^t.$$

Consequently, G is a composite test supermartingale for $\mathfrak{P}^{\mathcal{K}}$ which pointwise dominates E.

Proof. Set $E^{(0)} := E$. By Lemma 3.1, $E^{(0)}$ is bounded-stopping valid for $\mathfrak{P}^{\mathcal{K}}$, so Theorem 3.4 gives a `usa` Snell envelope $M^{(0)}$ satisfying $E^{(0)} \leq M^{(0)}$ pointwise, $\mathsf{T}_t M^{(0)}_{t+1} \leq M^{(0)}_t$ for every t, and $M^{(0)}_0 \leq 1$. Write $c := M^{(0)}_0$. By the supermartingale consequence of Theorem 3.4 and bounded optional sampling, $\mathbb{E}_{P^\kappa}[M^{(0)}_\tau] \leq c$ for every κ and $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. Proposition 4.2 therefore gives a Borel process $E^{(1)} := \mathsf{R}(M^{(0)})$ with $M^{(0)} \leq E^{(1)}$, $E^{(1)}_0 = c$, and $\mathbb{E}_{P^\kappa}[E^{(1)}_\tau] \leq c$ for every κ and $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. Thus $E^{(1)}$, viewed as a `usa` process, satisfies the bounded-stopping hypothesis of the scaled form of Theorem 3.4; we retain the sharper bound c throughout the iteration.

Proceed inductively. Suppose $E^{(n)}$ is Borel, $E^{(n)}_0 = c$, and $\mathbb{E}_{P^\kappa}[E^{(n)}_\tau] \leq c$ for every κ and $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. Apply the scaled form of Theorem 3.4. Its `usa` Snell envelope $M^{(n)}$ satisfies

$$E^{(n)} \leq M^{(n)}, \qquad \mathsf{T}_t M^{(n)}_{t+1} \leq M^{(n)}_t, \qquad M^{(n)}_0 \leq c.$$

Since $E^{(n)}_0 = c$ and $E^{(n)} \leq M^{(n)}$ pointwise, it follows that $M^{(n)}_0 = c$. Reflecting once more, define $E^{(n+1)} := \mathsf{R}(M^{(n)})$. Then $E^{(n+1)}$ is Borel, $M^{(n)} \leq E^{(n+1)}$ pointwise, $E^{(n+1)}_0 = c$, and $\mathbb{E}_{P^\kappa}[E^{(n+1)}_\tau] \leq c$ for every κ and $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. Thus the processes interlace pointwise as

$$E^{(0)} \leq M^{(0)} \leq E^{(1)} \leq M^{(1)} \leq E^{(2)} \leq \cdots.$$

For each $t \geq 0$ and $s \in \mathcal{Y}^t$, define

$$G_t(s) := \sup_{n \geq 0} E_t^{(n)}(s).$$

Every $E_t^{(n)}$ is Borel, so G_t is Borel as a countable supremum. The interlacing gives $G \geq E^{(0)} = E$ pointwise, and $G_0 = c \leq 1$. It remains to verify the one-step inequality. For every n and t,

$$E_t^{(n+1)} \geq M_t^{(n)} \geq \mathsf{T}_t M_{t+1}^{(n)} \geq \mathsf{T}_t E_{t+1}^{(n)},$$

where the final inequality uses monotonicity of T_t. Since $E_t^{(n)} \uparrow G_t$ pointwise, continuity from below of T_t gives

$$G_t = \sup_{n \geq 0} E_t^{(n+1)} \geq \sup_{n \geq 0} \mathsf{T}_t E_{t+1}^{(n)} = \mathsf{T}_t G_{t+1}.$$

Thus $\mathsf{T}_t G_{t+1}(s) \leq G_t(s)$ at every finite history. Finally, Lemma B.4 applied to the Borel process G shows that, under every $P^\kappa \in \mathfrak{P}^{\mathcal{K}}$,

$$\mathbb{E}_{P^\kappa}\left[G_{t+1}(Y_{1:t+1}) \,\middle|\, \mathcal{F}_t^{P^\kappa}\right] \leq G_t(Y_{1:t}) \qquad P^\kappa\text{-almost surely.}$$

Hence G is a composite test supermartingale for $\mathfrak{P}^{\mathcal{K}}$. □

Proposition 4.3 resolves the measurability obstruction left by § 3. More concretely, the final process is defined as a countable supremum of Borel reflected processes, which gives its Borel adaptedness, while the inequalities inherited through the interlaced Snell envelopes and continuity from below of T_t give its pointwise supermartingale property. Thus the two alternating operations contribute genuinely different parts of the final object. The argument is existential: measurable selection and reflection guarantee the required intermediate objects, but do not in general provide an algorithm for computing the dominating composite test supermartingale.

5 Complete-class theorem

The `usa` Snell construction (Protocol 2 and Theorem 3.4) and the Snell–reflection construction (Proposition 4.3) now give the main result. The assumptions are exactly those in § 2: $\mathcal{K}$ is nonempty-valued with analytic graph, and $\mathfrak{P}^{\mathcal{K}}$ contains all path laws generated by universally measurable $\mathcal{K}$-selections.

Theorem 5.1 (Complete class for analytic sequential hypotheses)**.** *Let $\mathcal{K} : \mathsf{H} \rightrightarrows \Delta(\mathcal{Y})$ be an analytic local model, and let $\mathfrak{P}^{\mathcal{K}}$ be the analytic sequential hypothesis it generates. Then composite test supermartingales form a complete class for $\mathfrak{P}^{\mathcal{K}}$ in the sense of Definition 2.7. More precisely, for every composite e-process $E = (E_t)_{t\geq 0}$ valid for $\mathfrak{P}^{\mathcal{K}}$ there exists a nonnegative Borel-adapted process $G = (G_t)_{t\geq 0}$ with $G_0 \leq 1$ such that*

$$G_t(s) \geq E_t(s) \qquad \text{for every } t \geq 0 \text{ and } s \in \mathcal{Y}^t, \tag{6}$$

and

$$\sup_{Q \in \mathcal{K}(s)} \int_{\mathcal{Y}} G_{t+1}(s \oplus y)\, Q(dy) \leq G_t(s) \qquad \text{for every } t \geq 0 \text{ and } s \in \mathcal{Y}^t. \tag{7}$$

In particular, G is a composite test supermartingale for $\mathfrak{P}^{\mathcal{K}}$ that pointwise dominates E.

Proof. Proposition 4.3 gives a nonnegative Borel process G with $G_0 \le 1$, satisfying the pointwise domination in (6) and the one-step inequalities in (7). Under any $P^\kappa \in \mathfrak{P}^{\mathcal{K}}$, the selected kernel $\kappa_t(Y_{1:t}, \cdot)$ is a version of the conditional law of Y_{t+1} given the past. Lemma B.4 therefore turns the pointwise kernel inequality into

$$\mathbb{E}_{P^\kappa}\left[G_{t+1}(Y_{1:t+1}) \,\middle|\, \mathcal{F}_t^{P^\kappa}\right] \le G_t(Y_{1:t}) \qquad P^\kappa\text{-almost surely.}$$

Thus G is a composite test supermartingale for every law in $\mathfrak{P}^{\mathcal{K}}$. □

Theorem 5.1 yields slightly more than the complete-class formulation alone records. The historywise inequality (7) is stronger than the complete-class statement itself. It holds after every finite history, including histories that may be null under some or all laws in the hypothesis. Accordingly, if a composite e-process crosses a rejection threshold on a given path, its dominating composite test supermartingale crosses no later.

6 Examples

We consider five examples. Several of the examples below have Borel graphs and, for suitably regular continuation values, admit Borel or even continuous upper transitions. These stronger conclusions use additional structure specific to the examples and should not be interpreted as a general consequence of the Borel-graph property; see Remark 3.5. Example 6.1 returns to the bounded conditional mean running example as a familiar reference point. Examples 6.2–6.4, covering conditional FOSD, sub-Gaussianity, and Wasserstein balls, impose distributional or tail restrictions that are statistically standard but are no longer described by finitely many conditional expectation constraints. Example 6.5 returns to conditional means in a different way: the exact mean restriction is relaxed to a small band precisely after histories compatible with an unobserved feasible object. Such compatibility conditions arise by projection and can naturally lead from Borel relations to analytic, non-Borel sets (Kline and Tamer, 2016; Beresteanu et al., 2012; Gu and Russell, 2023). The example therefore gives a genuinely analytic extension of the bounded-mean model while keeping each individual local set elementary.

Two structural checks recur across the examples below. First, many of the local classes are *not locally dominated*: there is no σ-finite measure that dominates every law in a local set $\mathcal{K}(s)$. A simple sufficient condition is the presence of uncountably many admissible laws carrying positive mass on pairwise disjoint measurable sets. Second, several of our local classes cannot be represented by the finite-dimensional one-step conditional constraint model underlying Clerico (2026),

$$\mathcal{H}_{\Phi,S} = \{Q : \mathbb{E}_Q\|\Phi\|_1 < \infty, \quad \mathbb{E}_Q[\Phi] \in S\}\,, \tag{8}$$

with a Borel feature map $\Phi = (\Phi_1, \dots, \Phi_d) : \mathcal{Y} \to \mathbb{R}^d$ and a nonempty convex set $S \subseteq \mathbb{R}^d$. The common argument for Examples 6.2–6.4 uses the one-step affine-domination result of Clerico (2026, Proposition 1). If such a null admitted a representation with a finite-dimensional feature map $\Phi = (\Phi_1, \dots, \Phi_d)$, then every one-step e-variable would be pointwise dominated by an affine function of Φ. Making an infinite family of e-variables simultaneously tight under a single null law then forces each member of the family to agree almost surely with its affine dominator, and hence to lie in the finite-dimensional span of $1, \Phi_1, \dots, \Phi_d$. The examples below construct tight families with arbitrarily large linearly independent subfamilies, yielding a contradiction. The analytic example is outside the same framework for a different, more immediate reason: each local section is itself a simple one-dimensional mean constraint, but which constraint applies depends on an analytic non-Borel set of histories. Closely related projection-based measurability issues arise in partial-identification problems when model compatibility is defined through the existence of a latent or structural parameter satisfying a measurable feasibility relation (Kline and Tamer, 2016; Beresteanu et al., 2012; Gu and Russell, 2023). The reusable obstruction lemmas are Propositions E.1 and E.2, with the Borel-graph obstruction in Proposition E.3; all example-specific proofs are collected in Appendix E.

These comparisons clarify the scope of Theorem 5.1. The finite-dimensional theory gives a much more explicit affine parametrization when it applies, while our local formulation also covers conditional nulls for which that parametrization is unavailable, including nondominated and genuinely analytic examples.

Example 6.1 (Bounded conditional means (running example))**.**
Composite null. Let $\mathcal{Y} = [0, 1]$ and fix $\mu \in (0, 1)$. For every history s, set

$$\mathcal{K}_\mu(s) := \{Q \in \Delta([0, 1]) : \mathbb{E}_Q[Y] = \mu\}.$$

The pointwise local specification implies that every $P \in \mathfrak{P}^{\mathcal{K}_\mu}$ satisfies

$$\mathbb{E}_P[Y_{t+1} \mid \mathcal{F}_t] = \mu \qquad P\text{-almost surely for every } t,$$

while allowing the rest of the conditional distribution to vary freely with time and history.

Graph regularity. The map $Q \mapsto \mathbb{E}_Q[Y]$ is continuous for weak convergence because $y \mapsto y$ is bounded and continuous on $[0, 1]$; see Lemma B.3. Thus $\mathcal{K}_\mu(s)$ is closed in the compact space $\Delta([0, 1])$, hence compact, and it is convex by linearity of expectation. Since the correspondence is independent of the history,

$$\mathrm{Gr}(\mathcal{K}_\mu) = \mathsf{H} \times \mathcal{K}_\mu,$$

so its graph is Borel. The same verification is recorded in Appendix E.2.

Not locally dominated. This can be seen directly. For each $x \in [0, \mu)$, define

$$Q_x := \frac{1-\mu}{1-x}\,\delta_x + \frac{\mu - x}{1-x}\,\delta_1.$$

Then $Q_x \in \mathcal{K}_\mu(s)$ and $Q_x(\{x\}) > 0$. If a σ-finite measure ν dominated every law in $\mathcal{K}_\mu(s)$, then $\nu(\{x\}) > 0$ for every $x \in [0, \mu)$. A σ-finite measure can have only countably many atoms of positive mass, giving a contradiction.

Finite-dimensional benchmark. The example lies inside the finite-dimensional framework of Clerico (2026): take $\Phi(y) = y$ and $S = \{\mu\}$. It therefore serves as a familiar finite-dimensional reference point for the examples below.

Betting and optimality. This setting is familiar for betting-based sequential inference. For predictable $\lambda_t \in [\frac{-1}{1-\mu}, \frac{1}{\mu}]$,

$$L_t(y) = 1 + \lambda_t(y - \mu)$$

has conditional expectation one under every admissible next-step law, and products of these factors are test martingales. Mixtures and adaptive choices of the stake give the usual bounded-mean betting constructions; see, for example, Howard et al. (2021) and related work on betting confidence sequences. For this particular null, Clerico (2025) prove a sharper complete-class result: every valid e-process is pointwise dominated by a coin-betting e-process of this form. Thus allowing arbitrary e-processes cannot improve on the coin-betting class for conditional mean testing. Our Theorem 5.1 recovers the same qualitative conclusion—that no testing power is gained by leaving the test-supermartingale class—without relying on the affine structure specific to the conditional-mean model.

Snell recursion. As discussed in § 3.1 and Protocol 1, the one-step upper expectation has the explicit form

$$(\mathsf{T}_t f)(s) = \operatorname{conc}\big(f(s \oplus \cdot)\big)(\mu)$$

for bounded continuous continuation values. For this model, the abstract Snell recursion reduces to a familiar concavification operation. In particular, for a continuous reward process, the finite-horizon Snell envelopes are continuous, hence Borel; the infinite-horizon envelope obtained by the monotone limit in Protocol 2 is therefore Borel as well. The continuity qualification is essential: the Borel-payoff construction in § 3.1, adapted from Frongillo (2025, Example 4.12), gives a non-Borel finite-horizon Snell envelope even though $\mathcal{K}_\mu$ has a compact Borel graph.

Beyond bounded means. The remaining examples are all locally nondominated. FOSD and sub-Gaussianity are also outside the finite-dimensional conditional-constraint framework without additional assumptions. For Wasserstein balls (Example 6.4), nondomination holds in the full compact-metric formulation, while the finite-dimensional question depends on the geometry of the center P_0; Appendix E.3 gives a simple diffuse-mass condition under which the ball is beyond the finite-dimensional framework. The final analytic example returns to mean constraints, but the active constraint changes with the history through an analytic non-Borel compatibility set arising from latent feasibility, so the correspondence itself has a genuinely non-Borel graph.

Example 6.2 (Conditional first-order stochastic dominance).
Composite null. At each time t, the observation is a pair

$$Y_t = (X_t, Z_t) \in [0,1]^2;$$

X_t and Z_t are the two coordinates observed in the same round, not observations at two different times. For a one-step law $Q \in \Delta([0,1]^2)$, let Q_X and Q_Z denote its two marginals. Define

$$\mathcal{K}_{\mathrm{FOSD}}(s) := \left\{ Q \in \Delta([0,1]^2) : Q_Z \preceq_{\mathrm{FOSD}} Q_X \right\},$$

independently of the history s. Equivalently,

$$F_{Q_X}(c) \le F_{Q_Z}(c) \qquad \text{for every } c \in [0,1].$$

The pointwise local restriction therefore implies that every $P \in \mathfrak{P}^{\mathcal{K}_{\mathrm{FOSD}}}$ satisfies, for every t,

$$P(X_{t+1} \le c \mid \mathcal{F}_t) \le P(Z_{t+1} \le c \mid \mathcal{F}_t) \qquad \text{for every } c \in [0,1],$$

P-almost surely. The conditional joint law may otherwise change arbitrarily with the past, and the within-round dependence between X_{t+1} and Z_{t+1} is unrestricted.

Graph regularity. The local set is closed, convex, and compact, so $\mathrm{Gr}(\mathcal{K}_{\mathrm{FOSD}})$ is Borel. Appendix E.2.1 derives these regularity claims from Lemma B.3.

Not locally dominated. The local class is not locally dominated. Appendix E.2.1 verifies this using Proposition E.1.

Infinite-dimensional. The local class has no finite-dimensional representation of the form $\mathcal{H}_{\Phi,S}$ in (8). For this obstruction, the useful witnesses are the threshold e-variables

$$E_c(x, z) = 1 + \eta\left(\mathbb{1}\{x \le c\} - \mathbb{1}\{z \le c\}\right), \qquad c \in (0, 1),$$

with $0 < \eta \le 1$. They are all tight under one null law with equal continuous marginals, and distinct thresholds give an infinite-dimensional family. Appendix E.2.1 formalizes the argument using Proposition E.2.

Betting and implications for testing. The same threshold functions furnish natural betting strategies for this null. Recent work develops anytime-valid betting tests for stochastic dominance using threshold information; for first-order stochastic dominance, Arnold et al. (2026) construct mixtures of asymptotically growth-rate-optimal e-variables and obtain a power-one test in their setting. Our conditional null uses the same basic distributional geometry but allows the admissible conditional law to change with the history, so this does not by itself identify an optimal predictable strategy here. Theorem 5.1 says that, for this conditional-FOSD hypothesis, every Borel composite e-process is pointwise dominated by a Borel composite test supermartingale. Thus improvements over a particular threshold strategy may require better bets or better predictable aggregation, but not a move outside the test-supermartingale class.

Snell regularity. For regular continuation values the Snell operator is also well behaved. Because the local set is constant and compact, if $f(s \oplus y)$ is jointly continuous then

$$(\mathsf{T}_t f)(s) = \sup_{Q \in \mathcal{K}_{\mathrm{FOSD}}} \mathbb{E}_Q[f(s \oplus Y)]$$

is continuous in s by the maximum theorem. Consequently, for a continuous reward process, the finite-horizon Snell envelopes are continuous, and the infinite-horizon envelope obtained as their monotone limit is Borel. For arbitrary Borel composite e-processes, however, we retain the general upper-semianalytic construction of § 3 and Protocol 2.

Example 6.3 (Conditional sub-Gaussianity)**.**
Composite null. Let $\mathcal{Y} = \mathbb{R}$ and normalize the sub-Gaussian variance proxy to one. Set, independently of the history s,

$$\mathcal{K}_{\mathrm{SG}}(s) := \left\{Q \in \Delta(\mathbb{R}) : \mathbb{E}_Q[e^{\lambda Y}] \le e^{\lambda^2/2} \quad \text{for every } \lambda \in \mathbb{R}\right\}.$$

The pointwise local restriction implies that every $P \in \mathfrak{P}^{\mathcal{K}_{\mathrm{SG}}}$ satisfies, for every t,

$$\mathbb{E}_P\left[e^{\lambda Y_{t+1}} \mid \mathcal{F}_t\right] \le e^{\lambda^2/2} \qquad \text{for every } \lambda \in \mathbb{R},$$

P-almost surely. A predictable location-and-scale version can be written by applying the same condition to a standardized increment.

Graph regularity. The local class $\mathcal{K}_{\mathrm{SG}}$ is convex and compact and has a Borel graph. The lower-semicontinuity and Prokhorov steps are recorded in Lemma B.3 and applied in Appendix E.2.2.

Not locally dominated. The local class is not locally dominated. Appendix E.2.2 gives the short verification using Proposition E.1.

Infinite-dimensional. Sub-Gaussianity is not representable by finitely many conditional expectations of a Borel feature map as in (8). The Gaussian law $\mathcal{N}(0,1)$ makes every

$$y \longmapsto e^{\lambda y - \lambda^2/2}, \qquad \lambda \in \mathbb{R},$$

tight, while these functions contain arbitrarily large linearly independent subfamilies. Appendix E.2.2 applies Proposition E.2 to formalize this saturation argument.

Betting and implications for testing. This is the usual sub-Gaussian assumption viewed as a null to be tested rather than merely as a condition used to prove concentration. The defining inequalities immediately yield the classical factors

$$L_\lambda(y) := \exp\left(\lambda y - \frac{\lambda^2}{2}\right),$$

so predictable choices of λ give the familiar composite test supermartingales of exponential form underlying time-uniform sub-Gaussian concentration (Howard et al., 2021). The one-step e-variable class is richer than this defining exponential family: Koolen et al. (2026) study log-growth-optimal e-variables against two-point alternatives and show that there is already nontrivial optimization to do at the one-step level. Our theorem does not say that the classical exponential bets are themselves optimal. Rather, it says that any testing advantage obtainable from a richer Borel composite e-process can be matched by some Borel composite test supermartingale, so the remaining problem is constructive—choosing and predictably aggregating good bets within that class.

Snell interpretation. For the classical exponential factors the robust one-step inequality is already exact under the Gaussian boundary law. Hence the corresponding exponential test martingales are fixed points of the Snell step in the supermartingale sense, so the Snell construction does not enlarge these explicit processes. For an arbitrary Borel composite e-process, however, this special structure is no longer available: the robust Snell envelope is treated in the general upper-semianalytic class, followed by the reflection and interlacing steps of § 4 to obtain a Borel dominating composite test supermartingale.

Example 6.4 (A conditional Wasserstein ball)**.**
Composite null. Let $(\mathcal{Y}, d)$ be an uncountable compact metric space, let $P_0 \in \Delta(\mathcal{Y})$ be a nominal law, write

$$D := \operatorname{diam}(\mathcal{Y}),$$

and fix $\rho \in (0, D)$. Define the local correspondence, independently of the history s, by

$$\mathcal{K}_{\mathrm{W}}(s) := \{Q \in \Delta(\mathcal{Y}) : W_1(Q, P_0) \le \rho\}.$$

The pointwise local restriction implies that every $P \in \mathfrak{P}^{\mathcal{K}_{\mathrm{W}}}$ satisfies, for every t,

$$W_1\left(\mathcal{L}_P(Y_{t+1} \mid \mathcal{F}_t), P_0\right) \le \rho \qquad P\text{-almost surely.}$$

The conditional distribution may therefore change arbitrarily over time, provided each admissible one-step kernel remains in the same transportation neighborhood of P_0. This is a natural model of bounded distribution shift.

Graph regularity. Because $W_1(\cdot, P_0)$ metrizes weak convergence on a compact metric space, $\mathcal{K}_{\mathrm{W}}$ is compact and convex and its graph is Borel. Appendix E.3 records this regularity step and its standard Wasserstein reference.

Not locally dominated. The local class is nondominated for every P_0 and every $\rho > 0$ in the uncountable compact setting. Appendix E.3 applies the common disjoint-support obstruction.

Finite-dimensionality. Whether a Wasserstein ball is beyond the finite-dimensional framework depends on its center. For example, if $P_0 = \delta_{y_0}$, then

$$W_1(Q, P_0) = \mathbb{E}_Q[d(Y, y_0)], \tag{9}$$

so the ball is a single expectation constraint. On the other hand, the usual continuous nominal models are genuinely richer. Appendix E.3 proves the following simple sufficient statement: on $\mathcal{Y} = [0, 1]$ with the Euclidean metric, if P_0 has a density bounded below by a positive constant on some nontrivial interval, then every sufficiently small positive-radius W_1 ball is not finitely feature-representable. The proof constructs one boundary law with infinitely many linearly independent tight Kantorovich–Rubinstein witnesses and then invokes the common saturation lemma.

Betting and implications for testing. The same duality gives a concrete test-supermartingale family. For every 1-Lipschitz f,

$$\mathbb{E}_Q[f] - \mathbb{E}_{P_0}[f] \le \rho \qquad \text{for every } Q \in \mathcal{K}_{\mathrm{W}}.$$

Hence

$$L_{f,\lambda}(y) := 1 + \lambda \left(f(y) - \mathbb{E}_{P_0}[f] - \rho \right), \qquad 0 \le \lambda \le \frac{1}{D + \rho},$$

is a nonnegative one-step e-variable. Predictable choices of the Lipschitz witness and the stake therefore generate composite test supermartingales. An exponential version follows just as directly from Hoeffding's lemma. These bets follow directly from Kantorovich–Rubinstein duality and are close in spirit to sequential betting based on learned witnesses for integral probability metrics (Shekhar and Ramdas, 2024). We have not found a previous reference that formulates, let alone characterizes an optimal procedure for, this conditional Wasserstein-ball null; related work instead studies Wasserstein uncertainty sets in robust fixed-sample testing and quickest change detection (Gao et al., 2018; Xie et al., 2024). Theorem 5.1 therefore narrows rather than solves the constructive problem: any testing advantage available to an arbitrary Borel composite e-process can be matched by a Borel composite test supermartingale, so one may search for power through better Lipschitz witnesses, stakes, or predictable aggregation without leaving that class.

Snell and distributionally robust optimization interpretation. The Snell operator has an independent optimization interpretation here. For a continuation payoff $g_s(y) := g(s \oplus y)$,

$$(\mathsf{T}_t g)(s) = \sup_{Q:\, W_1(Q, P_0) \le \rho} \mathbb{E}_Q[g_s].$$

For bounded continuous g_s, standard Wasserstein duality gives

$$(\mathsf{T}_t g)(s) = \inf_{\eta \ge 0} \left\{ \eta\rho + \int_{\mathcal{Y}} \sup_{y \in \mathcal{Y}} \big[g(s \oplus y) - \eta d(x, y) \big] P_0(dx) \right\};$$

see, for example, Esfahani and Kuhn (2018). Thus the dynamic primitive in Protocols 1 and 2 is the familiar worst-case expectation from Wasserstein distributionally robust optimization. Moreover, since the Wasserstein ball is compact and independent of the history, joint continuity of $(s, y) \mapsto g(s \oplus y)$ implies continuity of $s \mapsto (\mathsf{T}_t g)(s)$ by the maximum theorem. Hence continuous reward processes have continuous finite-horizon Snell envelopes, and the infinite-horizon envelope obtained as their monotone limit is Borel. For arbitrary Borel composite e-processes, however, the one-step upper expectation need not preserve Borel measurability, so the general upper-semianalytic Snell and Borelization machinery remains necessary.

Example 6.5 (A genuinely analytic extension of bounded conditional means)**.**
Motivation. Latent-variable and partial-identification models provide a natural statistical source of analytic admissibility conditions. A common structure is that an observed state or history is compatible with a model when there exists an unobserved parameter, latent variable, or structural object satisfying a measurable feasibility relation. Eliminating that latent witness is a projection operation, so even when the underlying relation is Borel, the resulting compatibility set need only be analytic. This issue appears explicitly in the partial-identification literature. For example, Kline and Tamer (2016, Remark 10) note that the set of reduced-form parameters for which an identified set is nonempty can fail to be Borel even when the criterion defining feasibility is measurable; related random-set approaches use measurable selections from sets of latent values compatible with the observations (Beresteanu et al., 2012; Gu and Russell, 2023).

A simple way for such a latent condition to enter a sequential mean model is for it to determine whether the next-step conditional mean is pinned down exactly or only known up to a small tolerance. This gives an analytic extension of the bounded conditional mean model used throughout the paper without changing the basic statistical primitive.

Composite null. Let $\mathcal{Y} = [0, 1]$, fix $\mu \in (0, 1)$, and choose tolerance

$$0 < \varepsilon < \min\{\mu, 1 - \mu\}.$$

Let Z be a Polish space and let $R \subseteq [0, 1] \times \mathsf{Z}$ be Borel with projection

$$C := \{y : \text{there exists } z \in \mathsf{Z} \text{ with } (y, z) \in R\}$$

analytic but non-Borel. One may read z as a latent feasibility certificate and C as the observed states compatible with at least one such certificate. At the root retain the bounded-mean null,

$$\mathcal{K}_{\mathrm{an}}(\varnothing) := \{Q \in \Delta([0, 1]) : \mathbb{E}_Q[Y] = \mu\}.$$

For a nonempty history $s = (y_1, \dots, y_t)$, set

$$\mathcal{K}_{\mathrm{an}}(s) := \begin{cases} \{Q \in \Delta([0, 1]) : |\mathbb{E}_Q[Y] - \mu| \le \varepsilon\}, & y_t \in C, \\ \{Q \in \Delta([0, 1]) : \mathbb{E}_Q[Y] = \mu\}, & y_t \notin C. \end{cases}$$

Thus the baseline exact conditional-mean restriction is relaxed to a band only after histories whose most recent observation is compatible with the latent model. Each local section remains a familiar compact convex mean-constraint set; the analytic complexity lies entirely in the history dependence of which section is active.

Graph regularity. Each value of the correspondence is compact and convex. Its graph, however, is analytic and genuinely non-Borel; Appendix E.4 gives the short verification.

Not locally dominated. Every local set contains the exact mean-μ class from Example 6.1, which is already locally nondominated. Hence $\mathcal{K}_{\mathrm{an}}$ is not locally dominated; Appendix E.4 records the reduction to Proposition E.1.

Beyond finite-dimensional conditional constraints. Although each individual section is described by a one-dimensional mean restriction, the correspondence as a whole cannot be represented by Clerico's fixed finite-dimensional conditional model, for a measurability reason rather than the saturation argument of Proposition E.2. More generally, Proposition E.3 shows that a local model generated by finitely many jointly Borel history-dependent features together with a Borel feasible-set correspondence has a Borel graph. The genuinely non-Borel graph of $\mathcal{K}_{\mathrm{an}}$ therefore rules out such a representation.

Snell and measurable-selection interpretation. The effect on the Snell construction is visible from the continuous continuation payoff $\varphi(y) := y$. For every nonempty history $s = (y_1, \ldots, y_t)$,

$$(\mathsf{T}_t\varphi)(s) = \sup_{Q\in\mathcal{K}_{\mathrm{an}}(s)} \mathbb{E}_Q[Y] = \mu + \varepsilon\mathbb{1}_C(y_t).$$

The input is continuous, while the output is upper semianalytic and non-Borel. Moreover, for any $\delta \in (0, \varepsilon/2)$, a δ-maximizing selector for this one-step problem need not admit a Borel version as a function of the history. A universally measurable selector does exist by Proposition C.4; Appendix E.4 verifies the non-Borel obstruction directly.

Complete-class consequence. In this example, every section is a simple conditional-mean model, like in our running example. However, the measurability of the history-to-model correspondence is now non-Borel. Theorem 5.1 nevertheless gives the same complete-class conclusion for this genuinely analytic null: every Borel composite e-process for $\mathfrak{P}^{\mathcal{K}_{\mathrm{an}}}$ is pointwise dominated by a Borel composite test supermartingale. The Borelization argument of § 4 is what allows the dominating process to be chosen Borel adapted despite the non-Borel intermediate Snell objects.

Summary. Together, the five examples range from familiar finite-dimensional restrictions to nondominated, infinite-dimensional, and genuinely analytic conditional nulls. Bounded means connect the local-model language to standard betting and sit inside the finite-dimensional theory. FOSD and sub-Gaussianity are natural, nondominated, infinite-dimensional conditional nulls for which useful composite test supermartingales already exist. Wasserstein balls give a flexible distributional-stability model that is nondominated in full generality and beyond finite-dimensional constraints under a mild diffuse-center condition, while also admitting a direct Kantorovich–Rubinstein betting construction. The final example returns to conditional means but lets a latent-feasibility condition determine whether the mean is fixed or lies in a small band; this produces a genuinely analytic non-Borel correspondence with simple compact sections and shows concretely why upper semianalytic dynamic programming and universally measurable selection can be necessary even for continuous continuation payoffs.

7 Discussion

We have shown that composite test supermartingales form a complete class for analytic sequential hypotheses: null families generated by measurable selections from an analytic correspondence of admissible one-step laws. The result does not require local domination, convexity, or compactness of the local sets. Its proof separates two issues that are often hidden in regular examples. The robust Snell envelope supplies the right pointwise supermartingale inequalities, while reflection and monotone interlacing recover Borel measurability without increasing the initial testing budget.

The examples in § 6 show that this scope includes both familiar and genuinely nonparametric conditional nulls. Bounded means recover the classical betting picture; FOSD and sub-Gaussianity lie beyond finite-feature conditional constraints; Wasserstein balls connect the upper transition operator to distributionally robust optimization; and the latent-feasibility extension shows how familiar conditional-mean restrictions can assemble into an analytic non-Borel local model.

These examples also separate structural completeness from the question of how explicitly the dominating process can be constructed. The main limitation of the general theorem is constructive. The measurable-selection and reflection steps are existential, so the dominating supermartingale supplied by the proof need not be directly computable from an arbitrary composite e-process. In regular examples the one-step upper expectation can be much more explicit—concavification for bounded means, or Wasserstein robust optimization for transport balls—and familiar composite test supermartingales may already be fixed points of the Snell operator. A natural next question is to identify conditions under which the complete-class result can be made operational: either through an explicit Snell operator, tractable approximate selectors, or a computable dominating supermartingale with controlled loss. Characterizing this constructive boundary is a natural direction for further work.

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A Path laws from universally measurable kernel selections

This appendix proves Lemma 2.2 and records the two standard measure-theoretic results used to pass from universally measurable one-step selections to a Borel probability law on path space.

Lemma A.1 (Borel modification under a fixed law)**.** *Let X and S be standard Borel spaces, let $\mu \in \Delta(X)$, and let $f : X \to S$ be universally measurable. Then there exists a Borel measurable map $\bar{f} : X \to S$ such that $f = \bar{f}$ μ-almost surely. In particular, if $f : X \to \Delta(\mathcal{Y})$ is universally measurable, it admits a Borel kernel-valued version under every fixed Borel probability law on X.*

This is the standard Borel-modification property of universally measurable maps; for the real-valued form see Bertsekas and Shreve (1996, Lemma 7.27), with the standard-Borel-valued statement obtained through a Borel isomorphism.

Theorem A.2 (Ionescu–Tulcea extension theorem)**.** *Let $S_1, S_2, \ldots$ be measurable spaces and let q_n be a probability kernel from $S_1 \times \cdots \times S_{n-1}$ to S_n for each $n \geq 1$. Then there exists a unique probability measure on $\prod_{n\geq 1} S_n$ whose finite-dimensional distributions are the iterated kernel products $q_1 \otimes \cdots \otimes q_n$. Equivalently, the coordinate process has q_n as a version of its nth one-step conditional law.*

This is the classical Ionescu–Tulcea construction; see, e.g., Kallenberg (2021).

Proof of Lemma 2.2. Fix a $\mathcal{K}$-selection $\kappa = (\kappa_t)_{t\geq 0}$. We recursively construct the finite-dimensional history laws. Set $\mu_0^\kappa := \delta_\varnothing$. Suppose μ_t^κ has been constructed on $\mathcal{Y}^t$. Since $\kappa_t : \mathcal{Y}^t \to \Delta(\mathcal{Y})$ is universally measurable and $\Delta(\mathcal{Y})$ is a standard Borel space, Lemma A.1 gives a Borel measurable map $\bar{\kappa}_t : \mathcal{Y}^t \to \Delta(\mathcal{Y})$ such that $\bar{\kappa}_t = \kappa_t$ μ_t^κ-almost surely. Define

$$\mu_{t+1}^\kappa(B) := \int_{\mathcal{Y}^t} \int_{\mathcal{Y}} \mathbb{1}_B(s \oplus y)\, \bar{\kappa}_t(s, dy)\, \mu_t^\kappa(ds), \qquad B \in \mathcal{B}(\mathcal{Y}^{t+1}).$$

Changing $\bar{\kappa}_t$ on a μ_t^κ-null set does not change μ_{t+1}^κ, so this recursion is independent of the chosen Borel version.

Applying Theorem A.2 to the Borel kernels $(\bar{\kappa}_t)_{t\geq 0}$ gives a unique Borel probability measure P^κ on $(\Omega, \mathcal{F})$ whose t-dimensional marginal is μ_t^κ. Because $\bar{\kappa}_t = \kappa_t$ μ_t^κ-almost surely, the usual iterated-integral formula may equivalently be written using the original universally measurable kernels κ_t. Moreover, $\bar{\kappa}_t(Y_{1:t}, \cdot)$ is a regular conditional distribution of Y_{t+1} given $\mathcal{F}_t$ under P^κ, and $\kappa_t(Y_{1:t}, \cdot)$ is an almost-surely equal version on the P^κ-completion of $\mathcal{F}_t$. Uniqueness follows because the finite-dimensional cylinder sets generate $\mathcal{F}$. □

B Measure-theoretic interface lemmas

B.1 Borel modification of universal stopping times

Lemma B.1 (Borel stopping-time modification)**.** *Let P be a Borel probability measure on Ω, and let $\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$. Then there exists a bounded stopping time σ with respect to the raw canonical filtration such that $P(\sigma = \tau) = 1$.*

Proof. Choose N such that $\tau \leq N$. For each $t \leq N$, write the exact-time event as $\{\tau = t\} = \{Y_{1:t} \in C_t\}$ for some $C_t \in \mathcal{U}_t$. Let P_t be the law of $Y_{1:t}$ under P. Because C_t belongs to the P_t-completion of $\mathcal{B}(\mathcal{Y}^t)$, choose a Borel set $D_t \subseteq \mathcal{Y}^t$ with $P_t(C_t \triangle D_t) = 0$. Set $\sigma := \inf\{0 \leq t \leq N : Y_{1:t} \in D_t\}$, with the convention that $\sigma = N$ if the set is empty. Then σ is a bounded Borel stopping time. Outside the P-null set $\bigcup_{t=0}^{N}\{Y_{1:t} \in C_t \triangle D_t\}$, membership in every exact-time region is unchanged, and hence $\sigma = \tau$. □

B.2 Borel parameterized integration

Lemma B.2 (Borel parameterized integral)**.** *Let X and Y be Polish spaces and let $h : X \times Y \to [0, \infty]$ be Borel. Then*

$$(x, Q) \longmapsto \int_Y h(x, y)\, Q(dy)$$

is Borel on $X \times \Delta(Y)$. The same conclusion holds for real-valued h on the Borel set where its positive and negative parts are both finite.

The result follows from the standard measurability theorem for kernels and integration, first for indicators and then by a monotone-class argument and monotone convergence; see, e.g., Kallenberg (2021).

B.3 Weak-topology facts for probability measures

Lemma B.3 (Weak-topology facts)**.** *Let S be Polish and equip $\Delta(S)$ with weak convergence. Then $\Delta(S)$ is Polish, and it is compact when S is compact. If $g : S \to \mathbb{R}$ is bounded and continuous, then $Q \mapsto \int g\, dQ$ is continuous. If $g : S \to [0, \infty]$ is lower semicontinuous, then $Q \mapsto \int g\, dQ$ is lower semicontinuous. If $F \subseteq S$ is closed, then*

$$\Delta(F) := \{Q \in \Delta(S) : Q(F) = 1\}$$

is weakly closed. Finally, a weakly closed uniformly tight family in $\Delta(S)$ is compact.

The continuity and lower-semicontinuity assertions are standard Portmanteau consequences, while the compactness statements are the corresponding Prokhorov facts; see, e.g., Kallenberg (2021). We use these facts only in the elementary forms stated above.

B.4 Conditional expectations of universally measurable continuation values

Lemma B.4 (Conditional-kernel identity)**.** *Fix a $\mathcal{K}$-selection κ and $t \geq 0$. Let $f : \mathcal{Y}^t \times \mathcal{Y} \to [0, \infty]$ be universally measurable. Then*

$$\mathbb{E}_{P^\kappa}\left[f(Y_{1:t}, Y_{t+1}) \,\middle|\, \mathcal{F}_t^{P^\kappa}\right] = \int_{\mathcal{Y}} f(Y_{1:t}, y)\, \kappa_t(Y_{1:t}, dy)$$

P^κ-almost surely, where $\mathcal{F}_t^{P^\kappa}$ denotes the P^κ-completion of $\mathcal{F}_t$.

Proof. We first treat bounded f. Let μ_t be the law of $Y_{1:t}$ under P^κ, and let $\bar{\kappa}_t$ be the Borel version of κ_t supplied by Lemma A.1. The joint law π of $(Y_{1:t}, Y_{t+1})$ is $\pi(ds, dy) = \mu_t(ds)\bar{\kappa}_t(s, dy)$. Since f is universally measurable, it is measurable with respect to the π-completion of $\mathcal{B}(\mathcal{Y}^t \times \mathcal{Y})$, so there is a bounded Borel function $\bar{f}$ satisfying $f = \bar{f}$ π-almost surely. The ordinary Borel-kernel identity gives

$$\mathbb{E}_{P^\kappa}\left[\bar{f}(Y_{1:t}, Y_{t+1}) \,\middle|\, \mathcal{F}_t^{P^\kappa}\right] = \int_{\mathcal{Y}} \bar{f}(Y_{1:t}, y)\, \bar{\kappa}_t(Y_{1:t}, dy)$$

P^κ-almost surely. The left-hand side is unchanged when $\bar{f}$ is replaced by f. On the right-hand side, Fubini–Tonelli applied to a Borel null superset of $\{f \neq \bar{f}\}$, together with $\kappa_t = \bar{\kappa}_t$ μ_t-almost surely, shows that

$$\int_{\mathcal{Y}} f(s, y)\, \kappa_t(s, dy) = \int_{\mathcal{Y}} \bar{f}(s, y)\, \bar{\kappa}_t(s, dy)$$

for μ_t-almost every s. This proves the bounded case. For general nonnegative f, apply the bounded result to $f \wedge m$ and use conditional monotone convergence. $\square$

B.5 Measurability of first contact

Lemma B.5 (First-contact stopping time). *Fix $N \in \mathbb{N}$ and let $E_t, M_t : \mathcal{Y}^t \to [0, \infty]$ be universally measurable for $t = 0, \ldots, N$, with $M_N = E_N$. Then*

$$\tau^* := \inf\{0 \le t \le N : M_t(Y_{1:t}) = E_t(Y_{1:t})\}$$

belongs to $\mathcal{T}^{\mathrm{u}}_{\mathrm{bd}}$.

Proof. For each $t \le N$, the contact set $C_t := \{s \in \mathcal{Y}^t : M_t(s) = E_t(s)\}$ belongs to $\mathcal{U}_t$ because E_t and M_t are universally measurable. For every $j \le N$, the event $\{\tau^* \le j\}$ is the union, over $t \le j$, of the pullbacks of C_t to $\mathcal{Y}^j$ under the corresponding prefix maps. Universal measurability is preserved under Borel preimages and finite unions, so $\{\tau^* \le j\}$ is represented by a set in $\mathcal{U}_j$. Finally, $M_N = E_N$ ensures $\tau^* \le N$. $\square$

C Analytic dynamic-programming facts

We collect the descriptive-set-theoretic and measurable-selection facts used by the Snell recursion. The first subsection summarizes the regularity classes that appear throughout the paper; the second records the specific projection, integration, and selection results used in the proofs.

C.1 Borel, analytic, and universally measurable classes

Let X be a Polish space. A subset of X is *Borel* if it belongs to the σ-field $\mathcal{B}(X)$ generated by the open sets. A set $A \subseteq X$ is *analytic* if there exist a Polish space Z and a Borel set $B \subseteq X \times Z$ such that

$$A = \mathrm{proj}_X(B).$$

Equivalently, analytic sets are the continuous images of Polish spaces. Every Borel set is analytic, but a projection of a Borel set need not be Borel; this is the classical phenomenon behind the genuinely non-Borel sets used in the paper.

A set $A \subseteq X$ is *universally measurable* if it belongs to

$$\mathcal{U}(X) := \bigcap_{P \in \Delta(X)} \mathcal{B}(X)^P,$$

where $\mathcal{B}(X)^P$ is the P-completion of the Borel σ-field. Every analytic set is universally measurable. Thus the hierarchy relevant here is

$$\{\text{Borel sets}\} \subseteq \{\text{analytic sets}\} \subseteq \{\text{universally measurable sets}\}.$$

The first inclusion can be strict because Borel projections may be analytic and non-Borel, while the second inclusion is the classical universal-measurability theorem for analytic sets. For standard references, see Kechris (1995, Sections 14–18 and Theorem 21.10); Bertsekas and Shreve (1996, Chapter 7) gives a formulation tailored to stochastic control and measurable selection.

There is a parallel hierarchy for functions. For an extended-real-valued function $f : X \to [-\infty, \infty]$, Borel measurability has its usual meaning. We call f *upper semianalytic* if

$$\{x \in X : f(x) > a\}$$

is analytic for every $a \in \mathbb{R}$, and *universally measurable* if it is measurable from $\mathcal{U}(X)$ to the Borel σ-field on $[-\infty, \infty]$. Consequently,

$$\{\text{Borel measurable functions}\} \subseteq \{\text{upper semianalytic functions}\} \subseteq \{\text{universally measurable functions}\}.$$

For indicators the set and function terminology agree in the expected way: $\mathbb{1}_A$ is upper semianalytic exactly when A is analytic. Upper semianalyticity is the natural convention for the suprema used below; the lower-semianalytic convention is obtained by reversing the inequalities, or equivalently by applying upper semianalyticity to $-f$.

The reason this hierarchy enters the proof is projection. If $h(s, Q)$ is upper semianalytic and $\mathrm{Gr}(\mathcal{K})$ is analytic, then

$$\left\{ s : \sup_{Q \in \mathcal{K}(s)} h(s, Q) > a \right\} = \mathrm{proj}_X \left(\mathrm{Gr}(\mathcal{K}) \cap \{(s, Q) : h(s, Q) > a\} \right),$$

with X denoting the history space under consideration. The set on the right is analytic, so the supremum is upper semianalytic, but it need not be Borel. At the same time, analytic graphs admit universally measurable selectors, whereas Borel selectors need not exist. This is the basic pattern behind the proof: the testing objects start in the Borel class, robust optimization produces upper-semianalytic value functions and universally measurable near-maximizers, and the reflection argument in § 4 returns the final dominating process to the Borel class.

C.2 Selection and upper-expectation facts

Lemma C.1 (Analytic sets and universal measurability)**.** *Let X and Y be Polish spaces. Every Borel subset of X is analytic, every analytic subset of X is universally measurable, and the projection of an analytic subset of $X \times Y$ onto X is analytic. Analytic sets are preserved under Borel preimages and under the finite unions and products used below. In particular, every upper semianalytic function is universally measurable.*

These are standard closure and measurability properties of analytic sets; see, e.g., Kechris (1995, Sections 14–18) and Bertsekas and Shreve (1996, Chapter 7).

Theorem C.2 (Jankov–von Neumann selection)**.** *Let X and A be standard Borel spaces and let $G \subseteq X \times A$ be analytic. There exists a universally measurable map $\phi : \mathrm{proj}_X(G) \to A$ such that $(x, \phi(x)) \in G$ for every $x \in \mathrm{proj}_X(G)$. Thus an analytic correspondence with nonempty sections admits a universally measurable selector.*

See Bertsekas and Shreve (1996, Proposition 7.49).

Lemma C.3 (Integration preserves upper semianalyticity)**.** *Let X and Y be standard Borel spaces and let $h : X \times Y \to [0, \infty]$ be upper semianalytic. Then*

$$(x, Q) \longmapsto \int_Y h(x, y)\, Q(dy)$$

is upper semianalytic on $X \times \Delta(Y)$.

This is the variable-measure form of the upper-semianalytic kernel-integration theorem; see Bertsekas and Shreve (1996, Proposition 7.48 and Corollary 7.48.1).

Proposition C.4 (Universally measurable near-maximizers)**.** *Let X and A be standard Borel spaces, let $G \subseteq X \times A$ be analytic with nonempty sections, and let $h : X \times A \to \mathbb{R}$ be bounded and upper semianalytic. Define*

$$v(x) := \sup\{h(x, a) : (x, a) \in G\}.$$

Then v is upper semianalytic and, for every $\delta > 0$, there exists a universally measurable selector $\phi_\delta : X \to A$ with $(x, \phi_\delta(x)) \in G$ and

$$h(x, \phi_\delta(x)) \geq v(x) - \delta$$

for every $x \in X$. No attainment of the supremum is required.

This is the measurable ε-selection theorem; see Bertsekas and Shreve (1996, Proposition 7.50).

Proof of Lemma 3.2. Define $I(s, Q) := \int_{\mathcal{Y}} f(s \oplus y)\, Q(dy)$ on $\mathcal{Y}^t \times \Delta(\mathcal{Y})$. Lemma C.3 implies that I is usa. Since $\mathrm{Gr}(\mathcal{K}) \cap (\mathcal{Y}^t \times \Delta(\mathcal{Y}))$ is analytic, for every $a \in \mathbb{R}$ the strict superlevel set of $\mathsf{T}_t f$ can be written as the projection

$$\{s : (\mathsf{T}_t f)(s) > a\} = \mathrm{proj}_{\mathcal{Y}^t} \left(\mathrm{Gr}(\mathcal{K}) \cap \{(s, Q) : I(s, Q) > a\}\right),$$

which is analytic by Lemma C.1. Thus $\mathsf{T}_t f$ is usa. Monotonicity is immediate from the definition. If $f_n \uparrow f$, monotone convergence gives, for each fixed s,

$$\sup_n (\mathsf{T}_t f_n)(s) = \sup_n \sup_{Q \in \mathcal{K}(s)} \int f_n(s \oplus y)\, Q(dy) = \sup_{Q \in \mathcal{K}(s)} \int f(s \oplus y)\, Q(dy) = (\mathsf{T}_t f)(s),$$

where the two suprema commute because both sides are the supremum over $\mathbb{N} \times \mathcal{K}(s)$. Finally, when f is bounded, Proposition C.4, applied to I on the analytic graph of $\mathcal{K}$, gives a universally measurable δ-maximizing selector κ_t^δ with the asserted inequality. $\square$

D Process reflection

This appendix proves Proposition 4.1. The argument uses a dual form of the First Reflection Theorem (Kechris, 1995, Theorem 35.10) together with a parameterized robust stopping problem.

Theorem D.1 (Dual First Reflection)**.** *Let X be Polish and let Φ be a property of subsets of X which is Π^1_1-on-Σ^1_1: for every Polish parameter space Z and every analytic set $A \subseteq Z \times X$, the set $\{z \in Z : \Phi(A_z)\}$ is coanalytic. If an analytic set $D \subseteq X$ has property Φ, then there is a Borel set $B \supseteq D$ which also has property Φ.*

We next verify the parameterized measurability needed to apply this theorem to the robust stopped-value cost. For analytic $D \subseteq Z \times \mathsf{H} \times [0,\infty)$ and $z \in Z$, write $D_z := \{(s,u) : (z,s,u) \in D\}$. For $m \in \mathbb{N}$ and $s \in \mathcal{Y}^t$, define the truncated section length

$$g^m_t(z,s) := \int_0^m \mathbb{1}_D(z,s,u)\,du.$$

Lemma D.2 (Parameterized section lengths)**.** *Let Z be Polish and let $D \subseteq Z \times \mathsf{H} \times [0,\infty)$ be analytic. For every $m \in \mathbb{N}$ and $t \geq 0$, the map $(z,s) \mapsto g^m_t(z,s)$ on $Z \times \mathcal{Y}^t$ is bounded and* `usa`*. Moreover, $g_{D_z}(s) = \sup_m g^m_t(z,s)$ for every $z \in Z$ and $s \in \mathcal{Y}^t$.*

Proof. The indicator of an analytic set is `usa` by Lemma C.1. Lemma C.3, applied to the normalized restriction of Lebesgue measure on $[0,m]$, shows that integrating this bounded `usa` function in u preserves upper semianalyticity; hence g^m_t is `usa` jointly in (z,s). The final identity follows from monotone convergence of the section measures as $m \to \infty$. □

Lemma D.3 (Parameterized robust stopped value)**.** *Let Z be Polish and let $D \subseteq Z \times \mathsf{H} \times [0,\infty)$ be analytic. Then the map $z \mapsto \mathcal{C}(D_z)$ is* `usa` *on Z.*

Proof. Fix $m \in \mathbb{N}$ and a finite horizon N. Starting from the bounded `usa` reward functions g^m_t, define the parameterized Snell recursion by $V^{m,N}_N(z,s) := g^m_N(z,s)$ and, for $t < N$,

$$C^{m,N}_t(z,s) := \sup_{Q \in \mathcal{K}(s)} \int_{\mathcal{Y}} V^{m,N}_{t+1}(z, s \oplus y)\,Q(dy), \qquad V^{m,N}_t(z,s) := g^m_t(z,s) \vee C^{m,N}_t(z,s).$$

Applying Lemma C.3, Lemma C.1, and the same projection argument as in the proof of Lemma 3.2 shows inductively that $V^{m,N}_t$ and $C^{m,N}_t$ are `usa` jointly in (z,s). In particular, $z \mapsto V^{m,N}_0(z,\varnothing)$ is `usa`.

For each fixed z, the usual finite-horizon robust stopping argument gives

$$V^{m,N}_0(z,\varnothing) = \sup_{\kappa} \sup_{\substack{\tau \in \mathcal{T}^{\mathrm{u}}_{\mathrm{bd}} \\ \tau \leq N}} \mathbb{E}_{P^\kappa}\big[g^m_\tau(z, Y_{1:\tau})\big].$$

For completeness, the upper bound follows because the Snell envelope is a bounded supermartingale under every P^κ and dominates the reward. For the reverse bound, Proposition C.4 gives universally measurable δ_t-maximizing kernels, and we stop in the universally measurable regions

$$\{s : g^m_t(z,s) \geq C^{m,N}_t(z,s) - \delta_t\},$$

with $\sum_{t<N} \delta_t$ arbitrarily small; the standard backward ε-optimality induction then gives the reverse inequality. Here z is fixed before the policy is selected, so no selector jointly measurable in (z,s) is required.

Every bounded stopping time is bounded by some deterministic horizon, and for fixed κ and τ, monotone convergence gives

$$\mathbb{E}_{P^\kappa}[g_{D_z}(Y_{1:\tau})] = \sup_m \mathbb{E}_{P^\kappa}[g^m_\tau(z, Y_{1:\tau})].$$

Consequently,

$$\mathcal{C}(D_z) = \sup_{N\geq 0} \sup_{m\geq 1} V_0^{m,N}(z, \varnothing).$$

This is a countable supremum of `usa` functions of z, and is therefore `usa`. $\square$

Proof of Proposition 4.1. Let $q := \mathcal{C}(D) < \infty$. For analytic subsets $S \subseteq \mathsf{H} \times [0, \infty)$, consider the property $\Phi_q(S)$ defined by $\mathcal{C}(S) \leq q$. To verify that this property is Π^1_1-on-Σ^1_1, let Z be Polish and $A \subseteq Z \times \mathsf{H} \times [0, \infty)$ be analytic. By Lemma D.3, the map $z \mapsto \mathcal{C}(A_z)$ is `usa`, so

$$\{z : \Phi_q(A_z)\} = \{z : \mathcal{C}(A_z) \leq q\}$$

is coanalytic. Theorem D.1 therefore yields a Borel set $B \supseteq D$ with $\mathcal{C}(B) \leq q$. Monotonicity of $\mathcal{C}$ under inclusion gives $q = \mathcal{C}(D) \leq \mathcal{C}(B)$, hence $\mathcal{C}(B) = \mathcal{C}(D)$. $\square$

E Technical details for the examples

The structural lemmas and technical verifications for § 6 are collected here. The weak-topology facts repeatedly used to verify closedness, compactness, and Borel graphs are stated earlier in Lemma B.3; here we record the reusable obstruction lemmas and then treat the examples in the same order as in § 6.

E.1 Reusable structural lemmas

Two structural criteria recur across the examples. The first rules out local domination. The second rules out finite-dimensional conditional-constraint representations.

Proposition E.1 (A nondomination criterion)**.** *Let $\mathcal{K}_0 \subseteq \Delta(\mathcal{Y})$. Suppose there is an uncountable index set A, laws $\{Q_\alpha : \alpha \in A\} \subseteq \mathcal{K}_0$, and pairwise disjoint measurable sets $\{B_\alpha : \alpha \in A\}$ such that*

$$Q_\alpha(B_\alpha) > 0 \qquad \textit{for every } \alpha \in A.$$

Then no σ-finite measure dominates every law in $\mathcal{K}_0$.

Proof. Suppose a σ-finite measure ν dominated every Q_α. Then $Q_\alpha(B_\alpha) > 0$ implies $\nu(B_\alpha) > 0$ for every α. Write $\mathcal{Y} = \bigcup_{n\geq 1} A_n$ with $\nu(A_n) < \infty$. For every α, some n satisfies $\nu(B_\alpha \cap A_n) > 0$. For fixed n, the pairwise disjoint sets $B_\alpha \cap A_n$ can have positive ν-mass for at most countably many α. Taking the countable union over n shows that only countably many B_α can have positive mass, contradicting uncountability of A. $\square$

The special case used in Appendices E.2, E.2.2, E.3, and E.4 takes B_α to be distinct singletons.

Proposition E.2 (Saturating e-variables obstruct finite-dimensional representations)**.** *Let $\mathcal{K}_0 \subseteq \Delta(\mathcal{Y})$ be a one-step null on a standard Borel space. Suppose there is a law $Q_0 \in \mathcal{K}_0$ and a family of nonnegative Borel functions $\{E_\alpha : \alpha \in A\}$ such that*

$$\sup_{Q\in\mathcal{K}_0} \mathbb{E}_Q[E_\alpha] \leq 1, \qquad \mathbb{E}_{Q_0}[E_\alpha] = 1$$

for every α, and such that the span of $\{E_\alpha : \alpha \in A\}$ is infinite-dimensional modulo Q_0-almost-sure equality. Then there do not exist a finite d, a Borel map $\Phi : \mathcal{Y} \to \mathbb{R}^d$, and a nonempty convex set $S \subseteq \mathbb{R}^d$ for which

$$\mathcal{K}_0 = \{Q : \mathbb{E}_Q\|\Phi\|_1 < \infty, \quad \mathbb{E}_Q[\Phi] \in S\}.$$

Proof. Assume such a representation exists. For each α, the one-step affine-domination result of Clerico (2026) gives a valid e-variable $\widehat{E}_\alpha$ dominating E_α pointwise on the effective domain and having the form

$$\widehat{E}_\alpha = a_\alpha + b_\alpha^\top \Phi.$$

Since Q_0 belongs to the null,

$$1 = \mathbb{E}_{Q_0}[E_\alpha] \le \mathbb{E}_{Q_0}[\widehat{E}_\alpha] \le 1.$$

Hence $\widehat{E}_\alpha - E_\alpha \ge 0$ has zero Q_0-expectation, so

$$E_\alpha = \widehat{E}_\alpha \qquad Q_0\text{-almost surely.}$$

Thus every E_α belongs, modulo Q_0-null sets, to

$$\operatorname{span}\{1, \Phi_1, \dots, \Phi_d\},$$

a vector space of dimension at most $d+1$. This contradicts the assumed infinite-dimensionality of the witness family. □

Proposition E.2 is the common argument used for FOSD (Appendix E.2.1), sub-Gaussianity (Appendix E.2.2), and the nondegenerate Wasserstein balls (Appendix E.3). The analytic example uses a different obstruction.

Proposition E.3 (Borel finite-feature models have Borel graphs)**.** *Let $\Phi : \mathsf{H} \times \mathcal{Y} \to \mathbb{R}^d$ be Borel, and let $S : \mathsf{H} \rightrightarrows \mathbb{R}^d$ have Borel graph. Define*

$$\mathcal{K}_{\Phi,S}(s) := \left\{ Q \in \Delta(\mathcal{Y}) : \int \|\Phi(s,y)\|_1 \, Q(dy) < \infty, \quad \int \Phi(s,y)\, Q(dy) \in S(s) \right\}.$$

Then $\mathrm{Gr}(\mathcal{K}_{\Phi,S})$ *is Borel in* $\mathsf{H} \times \Delta(\mathcal{Y})$*. In particular, a local correspondence with genuinely non-Borel graph cannot admit such a representation.*

Proof. By Lemma B.2, the integrals of the positive and negative parts of each coordinate of Φ, as well as the integral of $\|\Phi\|_1$, are Borel functions of (s, Q). Hence the integrability set is Borel and the vector integral is Borel on that set. Membership of the integral in $S(s)$ is then the inverse image of $\mathrm{Gr}(S)$ under a Borel map. Thus $\mathrm{Gr}(\mathcal{K}_{\Phi,S})$ is Borel. □

E.2 Bounded conditional means

By Lemma B.3, the map

$$Q \longmapsto \int y \, Q(dy)$$

is continuous under weak convergence on $\Delta([0,1])$, and $\Delta([0,1])$ is compact. Hence the mean-μ level set is closed and compact, and the graph of the constant correspondence is Borel.

For nondomination, for each $x \in [0, \mu)$ define

$$p_x := \frac{1-\mu}{1-x}, \qquad Q_x := p_x \delta_x + (1-p_x)\delta_1.$$

Then $Q_x \in \mathcal{K}_\mu$ and $Q_x(\{x\}) = p_x > 0$. The singletons $\{x\}$ are pairwise disjoint for uncountably many x, so Proposition E.1 applies.

For completeness, the concave-envelope identity used in § 3.1 can be written as

$$\sup_{Q:\, \mathbb{E}_Q Y = \mu} \mathbb{E}_Q[f(Y)] = \mathrm{conc}(f)(\mu)$$

for bounded continuous $f : [0,1] \to \mathbb{R}$. The upper bound follows from Jensen's inequality applied to every concave majorant. For the reverse inequality, the least concave majorant at μ is the supremum of chord values

$$\frac{b-\mu}{b-a} f(a) + \frac{\mu-a}{b-a} f(b), \qquad 0 \le a \le \mu \le b \le 1,$$

with the obvious interpretation when $a = b = \mu$. The corresponding two-point distribution has mean μ.

E.2.1 Conditional first-order stochastic dominance

For $Q \in \Delta([0,1]^2)$,

$$Q_Z \preceq_{\mathrm{FOSD}} Q_X$$

is equivalent to

$$\int \phi \, dQ_X \ge \int \phi \, dQ_Z$$

for every bounded continuous increasing $\phi : [0,1] \to \mathbb{R}$. By Lemma B.3, each inequality defines a weakly closed half-space, so $\mathcal{K}_{\mathrm{FOSD}}$ is closed and convex. The same lemma gives compactness because $[0,1]^2$ is compact. Hence the constant correspondence has Borel graph.

For nondomination, every point mass $\delta_{(x,z)}$ with $x \ge z$ belongs to the null. The uncountable family of singletons in the triangle

$$\{(x,z) \in [0,1]^2 : x \ge z\}$$

therefore satisfies Proposition E.1.

We next verify the hypotheses of Proposition E.2. Fix $0 < \eta \le 1$ and, for $c \in (0,1)$, define

$$E_c(x,z) := 1 + \eta \left(\mathbb{1}\{x \le c\} - \mathbb{1}\{z \le c\} \right).$$

The factor is nonnegative, and for every $Q \in \mathcal{K}_{\mathrm{FOSD}}$,

$$\mathbb{E}_Q[E_c] = 1 + \eta\big(F_{Q_X}(c) - F_{Q_Z}(c)\big) \le 1.$$

Let

$$Q_0 := \mathrm{Unif}[0,1] \otimes \mathrm{Unif}[0,1].$$

Its marginals agree, so Q_0 is in the null and $\mathbb{E}_{Q_0}[E_c] = 1$ for every c.

It remains to show infinite-dimensionality modulo Q_0. Take distinct $c_1, \dots, c_n$ and suppose

$$\sum_{j=1}^{n} a_j E_{c_j}(x, z) = 0$$

for Lebesgue-almost every (x, z). Set

$$g(u) := \sum_{j=1}^{n} a_j \mathbb{1}\{u \le c_j\}.$$

Then

$$\sum_{j=1}^{n} a_j + \eta\big(g(x) - g(z)\big) = 0$$

for almost every (x, z). Interchanging x and z and adding gives $\sum_j a_j = 0$, after which $g(x) = g(z)$ for almost every pair. Thus g is almost everywhere constant. It vanishes above $\max_j c_j$, so it is zero almost everywhere. The jump at each distinct c_j has size $-a_j$, hence every $a_j = 0$. Proposition E.2 now shows that $\mathcal{K}_{\text{FOSD}}$ has no finite-dimensional representation of Clerico's form.

E.2.2 Conditional sub-Gaussianity

For fixed $\lambda \in \mathbb{R}$, Lemma B.3 gives lower semicontinuity of

$$Q \longmapsto \int e^{\lambda y}\, Q(dy)$$

under weak convergence. Hence the sublevel set at $e^{\lambda^2/2}$ is closed, and arbitrary intersection over λ shows that $\mathcal{K}_{\text{SG}}$ is closed. It is convex by linearity of expectation. Moreover, the MGF bound gives the uniform tail estimate

$$Q(|Y| \ge r) \le 2e^{-r^2/2}, \qquad Q \in \mathcal{K}_{\text{SG}},$$

so the class is uniformly tight. The Prokhorov part of Lemma B.3 therefore gives compactness, and the constant local correspondence has Borel graph.

For nondomination, for every $x \in (0, 1]$ let

$$Q_x := \frac{1}{2}\delta_{-x} + \frac{1}{2}\delta_x.$$

Then

$$\mathbb{E}_{Q_x}[e^{\lambda Y}] = \cosh(\lambda x) \le e^{\lambda^2 x^2/2} \le e^{\lambda^2/2},$$

so $Q_x \in \mathcal{K}_{\text{SG}}$. The singletons $\{x\}$ give the disjoint sets required by Proposition E.1.

For the finite-dimensional obstruction, define

$$E_\lambda(y) := \exp\left(\lambda y - \frac{\lambda^2}{2}\right).$$

These are e-variables for $\mathcal{K}_{\text{SG}}$ by definition. Let $\gamma = N(0, 1)$. Then $\gamma \in \mathcal{K}_{\text{SG}}$ and

$$\mathbb{E}_\gamma[E_\lambda] = 1$$

for every λ. To see linear independence, take distinct $\lambda_1, \dots, \lambda_n$ and suppose

$$\sum_{j=1}^{n} a_j e^{\lambda_j y - \lambda_j^2/2} = 0$$

for γ-almost every y. The left side is continuous and γ has a strictly positive density, so the identity holds for every y. Absorb the constants $e^{-\lambda_j^2/2}$ into the coefficients and differentiate $k = 0, \dots, n-1$ times at $y = 0$. The resulting Vandermonde system is nonsingular because the λ_j are distinct. Thus all coefficients vanish. Proposition E.2 applies.

E.3 Conditional Wasserstein balls

Regularity and nondomination. On a compact metric space, W_1 metrizes weak convergence; see, e.g., the Wasserstein preliminaries in Esfahani and Kuhn (2018). Hence

$$\mathcal{K}_{\mathrm{W}} = \{Q : W_1(Q, P_0) \le \rho\}$$

is weakly closed and therefore compact by Lemma B.3; it is convex by convexity of $W_1(\cdot, P_0)$. The graph of the constant correspondence is Borel.

For nondomination, fix

$$0 < \varepsilon < \frac{\rho}{D}$$

and, for $x \in \mathcal{Y}$, set

$$Q_x := (1-\varepsilon)P_0 + \varepsilon\delta_x.$$

Coupling the common $(1-\varepsilon)P_0$ part identically and moving only the remaining mass gives

$$W_1(Q_x, P_0) \le \varepsilon D < \rho.$$

A probability measure has at most countably many atoms, so there is an uncountable set $A \subseteq \mathcal{Y}$ on which $P_0(\{x\}) = 0$. For every $x \in A$,

$$Q_x(\{x\}) = \varepsilon > 0.$$

Proposition E.1 applies to the singletons $\{x\}$.

A clean sufficient condition for being beyond finite-dimensional constraints. We give a concrete statement in the standard one-dimensional setting. Let $\mathcal{Y} = [0, 1]$ with $d(x, y) = |x - y|$. Suppose there are an interval $I = [a, b]$ of length $L = b - a > 0$ and a constant $c > 0$ such that

$$P_0(A) \ge c\,\mathrm{Leb}(A) \qquad \text{for every Borel } A \subseteq I.$$

In particular, this holds if P_0 has a density bounded below by a positive constant on a nontrivial interval. We show that if

$$0 < \rho < \frac{cL^2}{16},$$

then $\mathcal{K}_{\mathrm{W}}$ is not finitely feature-representable.

Set $\ell = L/4$ and define adjacent intervals

$$I_1 = [a, a+\ell], \qquad I_2 = [a+\ell, a+2\ell].$$

Let

$$\alpha := \frac{\rho}{\ell^2} < c,$$

and define the probability measure

$$Q_\star := P_0 - \alpha \ \mathrm{Leb}|_{I_1} + \alpha \ \mathrm{Leb}|_{I_2}.$$

The lower bound on P_0 ensures nonnegativity on I_1. Using the one-dimensional formula

$$W_1(P, Q) = \int_0^1 |F_P(u) - F_Q(u)| \, du,$$

the CDF difference generated by this transfer is a triangular excursion over $I_1 \cup I_2$ with area $\alpha\ell^2$. Hence

$$W_1(Q_\star, P_0) = \rho.$$

Choose a 1-Lipschitz function f_0 that has slope 1 on $I_1 \cup I_2$ and is constant on $[a + 2\ell, b]$. For example, after adding an irrelevant constant, one may take

$$f_0(y) = \min\{\max\{y - a, 0\}, 2\ell\}.$$

A direct calculation gives

$$\mathbb{E}_{Q_\star}[f_0] - \mathbb{E}_{P_0}[f_0] = \rho,$$

so f_0 is a Kantorovich–Rubinstein witness for the boundary pair.

There is still a nontrivial interval

$$J := (a + 2\ell, b)$$

on which $Q_\star = P_0$ and f_0 is constant. Choose pairwise disjoint nonempty open intervals $J_k \subset J$ and nonzero tent functions h_k supported on J_k with Lipschitz constant at most $1/2$. After scaling the tents if necessary, the functions

$$f_k := f_0 + h_k$$

are 1-Lipschitz. Because $Q_\star - P_0$ is supported on $I_1 \cup I_2$,

$$\mathbb{E}_{Q_\star}[f_k] - \mathbb{E}_{P_0}[f_k] = \rho$$

for every k.

Fix a sufficiently small common $\lambda > 0$ and define

$$E_k(y) := 1 + \lambda \left(f_k(y) - \mathbb{E}_{P_0}[f_k] - \rho \right).$$

Since $[0, 1]$ is compact and the f_k are uniformly bounded after the common normalization, λ can be chosen so that all E_k are nonnegative. Kantorovich–Rubinstein duality gives, for every $Q \in \mathcal{K}_{\mathrm{W}}$,

$$\mathbb{E}_Q[E_k] \leq 1,$$

while at the boundary law

$$\mathbb{E}_{Q_\star}[E_k] = 1.$$

Finally, the family $\{E_k\}$ is linearly independent modulo $Q_\star$. Indeed, on J we have $Q_\star = P_0 \geq c$ Leb and the supports of the h_k are disjoint. For any finite relation $\sum_{j=1}^m a_j E_{k_j} = 0$ $Q_\star$-almost surely, first evaluate on a positive-measure subset of J outside all of the bump supports to obtain the common baseline relation. Then restrict to J_{k_j} and subtract that baseline relation; this leaves

$$a_j \lambda h_{k_j} = 0$$

almost everywhere on J_{k_j}, so $a_j = 0$. Proposition E.2 completes the proof.

The diffuse-center condition used in Appendix E.3 is only sufficient, not necessary. It is included to make precise a broad natural regime in which the Wasserstein ball is beyond the finite-dimensional framework. By (9), a ball centered at a point mass can instead reduce to a single expectation constraint.

E.4 Analytic extension of the bounded-mean model

For $t \geq 1$, let

$$C_t := \{s = (y_1, \ldots, y_t) : y_t \in C\}.$$

The last-coordinate map is continuous, so Lemma C.1 implies that C_t is analytic. It is non-Borel whenever C is: if C_t were Borel, its preimage under the continuous embedding $y \mapsto (0, \ldots, 0, y)$ would make C Borel.

Write

$$\mathcal{K}^{=} := \{Q \in \Delta([0,1]) : \mathbb{E}_Q[Y] = \mu\}, \qquad \mathcal{K}^{\text{band}} := \{Q \in \Delta([0,1]) : |\mathbb{E}_Q[Y] - \mu| \leq \varepsilon\}.$$

Both sets are convex and weakly closed, hence compact, by Lemma B.3 and compactness of $\Delta([0,1])$. Moreover $\mathcal{K}^{=} \subseteq \mathcal{K}^{\text{band}}$. For $t \geq 1$, the graph of the time-t correspondence is

$$\mathrm{Gr}_t(\mathcal{K}_{\text{an}}) = \big(\mathcal{Y}^t \times \mathcal{K}^{=}\big) \cup \big(C_t \times \mathcal{K}^{\text{band}}\big).$$

The first term is Borel and the second is analytic, so Lemma C.1 shows that the graph is analytic. It is not Borel. Indeed, because $\mu + \varepsilon < 1$, the fixed law

$$Q^* := \delta_{\mu+\varepsilon}$$

belongs to $\mathcal{K}^{\text{band}} \setminus \mathcal{K}^{=}$, and therefore

$$\{s : (s, Q^*) \in \mathrm{Gr}_t(\mathcal{K}_{\text{an}})\} = C_t.$$

A Borel graph would make this section Borel, contradicting the choice of C.

Local nondomination follows because every $\mathcal{K}_{\text{an}}(s)$ contains $\mathcal{K}^{=}$. The exact mean-μ class contains the uncountable family of two-point laws used in Appendix E.2, so Proposition E.1 applies at every history. Proposition E.3 also shows that this genuinely non-Borel correspondence cannot be generated by a finite-dimensional jointly Borel feature map and a Borel feasible-set correspondence. In particular it is outside the fixed finite-dimensional conditional model of Clerico (2026).

Finally, take the continuous payoff $\varphi(y) = y$. For $s \in \mathcal{Y}^t$ with $t \geq 1$,

$$(\mathsf{T}_t\varphi)(s) = \begin{cases} \mu + \varepsilon, & s \in C_t, \\ \mu, & s \notin C_t, \end{cases} = \mu + \varepsilon \mathbb{1}_{C_t}(s).$$

Thus the upper transition is upper semianalytic and non-Borel even though the continuation payoff is continuous and every section of the local model is compact and convex.

To see that approximate maximizers may also require universal measurability, fix $\delta \in (0, \varepsilon/2)$ and let κ_t be a δ-optimal selector. If $s \in C_t$, then

$$\int y\,\kappa_t(s, dy) \geq \mu + \varepsilon - \delta > \mu + \frac{\varepsilon}{2},$$

whereas if $s \notin C_t$, every admissible law has mean exactly μ. Hence

$$C_t = \left\{ s : \int y\,\kappa_t(s, dy) > \mu + \frac{\varepsilon}{2} \right\}.$$

The map $Q \mapsto \int y\,Q(dy)$ is continuous, so

$$U := \left\{ Q : \int y\,Q(dy) > \mu + \frac{\varepsilon}{2} \right\}$$

is open in $\Delta(\mathcal{Y})$. If κ_t were Borel, then $C_t = \kappa_t^{-1}(U)$ would be Borel, a contradiction. A universally measurable δ-maximizing selector exists by Proposition C.4.